

WEAK LEARNING FAILURE ALONE IS NOT GRADIENT STARVATION: EXACT FINITE-WIDTH DYNAMICS FOR A PAIRED CAUSAL TEST

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Paper under double-blind review

ABSTRACT

A weak feature can remain poorly learned for several different reasons: its weak-only trajectory may not reach a specified target within the observation horizon, a strong cue may change cross-entropy weighting, response-specific parameter geometry may suppress its update, or earlier optimization help may not yet have been repaid by later suppression. Weak learning failure alone is not gradient starvation. We therefore compare a both-feature trajectory with a same-initialization weak-only trajectory in which only the strong input channel is ablated. This paired intervention distinguishes relative transfer, rate suppression, outcome suppression, weak-only nonlearnability under the specified target and horizon, and causal starvation. A failed weak-only first-hit gate licenses neither a starvation certificate nor evidence that starvation is absent; the causal status is indeterminate under that gate.

For full-batch logistic gradient flow, the exact finite-width weak-response dynamics are $\dot{M}_w = c_w^\top q(r)/n$, where $q_i(r) = \sigma(-r_i)$ is the cross-entropy weighting of signed logit r_i and $c_w = J\nabla M_w$ is the response-specific response-logit geometry. The companion signed-logit flow is $\dot{r} = \Theta q(r)/n$. These chain-rule identities require no dynamical mean-field theory (DMFT) or response-coordinate closure. Freezing (Θ, c_w) defines a nonlinear logistic tangent surrogate; continuous Grönwall and experiment-matched discrete secant-kernel bounds compare it with nonlinear training only under supplied kernel-movement control. Pathwise, for arbitrary sign changes, $\Delta = P - N$: outcome suppression occurs exactly when accumulated negative relative drift overtakes accumulated positive relative drift. Causal starvation additionally requires independently demonstrated weak-only learnability. A noiseless rank-one log drift ratio gives sufficient conditions for one rate crossing. In the dense-linear model, the realized initialization gives an exact pre-trajectory $d(0)$ threshold and an existential strict local outcome sign outside a caller indifference band; where proved, local jet information such as $d'(0)$ supplies further assumption-scoped local signs. Uniform product-ball bounds, when independently proved, imply crossing, outcome, nondegenerate learnability, or finite-horizon safety, but no such nonvacuous general positive-lag constants are currently constructed. Thus stronger global parameter-level prediction from ρ , lag, and recurrent geometry remains unresolved. We also prove stability of quantitatively isolated transverse first hits under uniform- error, radius, and mesh hypotheses and an exact six-scalar zero-lag wide limit for any finite realized recurrent matrix whose noiseless symmetric reference is suppressed from initialization while its weak-only trajectory remains learnable. As local mechanism-level control, in the fixed implemented chart, feasible uncapped target-attaining Counterfactual Drift Correction (CDC) is the unique minimum-Euclidean-norm correction satisfying zero instantaneous strong-drift change and the weak-drift lower bound; its finite-step guarantee is local and one-step.

In controlled dense-linear E1, all 32 ρ -seed cells per lag at lags 0, 2, and 4 (four strengths by eight seed IDs) have positive paired weak-AUC deficits and learnable weak-only controls; at each fixed lag, the mean deficit increases over the measured ρ grid. All 32 lag-8 cells fail the weak-only gate and are causally indeterminate

under $\beta = 0.25, H = 5$. At the fixed $(\rho, \text{lag}) = (4, 2)$ nonlinear cell, tanh has complete transfer-to-starvation certificates in 3/8 observed seeds; GRU has drift and response crossings in 5/8 but 0/8 causal certificates because every weak-only path fails $\beta = 0.5$. A post-hoc terminal sensitivity audit, derived without retraining, places tanh $B_W(H)$ in $[4.413, 4.676]$ and GRU in $[0.0828, 0.187]$; it is not a replacement for the first-hit gate or an architecture ranking. A sealed five-endpoint frozen- kernel E19 pilot fails its preregistered rule. Only after that failure, E20's fresh restricted 4×8 data/model-seed factorial classifies response-crossing events in 64/64 records and drift-crossing events in 60/64, with two-way-bootstrap 95% lower endpoints 1.0 and 0.8125. These are one-cell event-classification results, not phase, causal, learnability, calibrated-time, magnitude, prevalence, or kernel-stability validation. E20 remains the submission-primary empirical study with only its frozen drift/response-crossing endpoints primary; later clean-source extensions do not reprioritize it. E26 has 64 semi-real generated-cue records: MNIST 3-vs-8 and FashionMNIST 0-vs-6 each have 0/32 weak-only-learnable and 0/32 causal records, with mean weak-AUC gaps $-1.2362842136667493 \times 10^{-3}$ (95% CI $[-3.4405428466004646, 1.0159591371180453] \times 10^{-3}$) and $-1.0921728159018996 \times 10^{-4}$ (95% CI $[-4.129384520886106, 3.182670049955049] \times 10^{-3}$), respectively; this is negative/indeterminate. The failed causal gate blocks a false-positive starvation interpretation but licenses no absence claim. In E27's 192 records, tanh's all- β estimate is 0.2916666666666667 (95% CI $[0.16666666666666666, 0.4166666666666667]$) and GRU's is 0 (95% CI $[0, 0]$), with Holm-adjusted $p = 1$ for each; the frozen endpoint is negative and supports no architecture ranking. E28's 256 method-records pass the frozen joint weak-rescue/strong-preservation tradeoff only against the non-canonical Bloop-style shadow-target rescue; CDC does not beat the non-canonical PCGrad-style shadow- target rescue under the frozen joint criterion. No SOTA, canonical parity, or broad CDC superiority follows. The resulting chain is exact dynamics $\rightarrow$ matched causal intervention $\rightarrow$ pathwise attribution $\rightarrow$ local mechanism-level control. A general positive-lag optimization-time DMFT remains open.

1 INTRODUCTION

A weak predictive feature can remain small for several distinct reasons. The weak-only task may not reach a chosen target within the observation horizon; a strong cue may reduce cross-entropy weight on already-correct samples; the response-specific parameter geometry may route little gradient toward the weak response; or a later period of rate suppression may be too short to erase earlier help. Cross-entropy can indeed favor one predictive feature strongly enough to starve another (Pezeshki et al., 2021), but weak learning failure alone is not gradient starvation. Recurrent networks make the diagnosis especially delicate because cue strength, lag, shared recurrent parameters, and evolving response geometry all affect the update.

We make the diagnosis with a matched intervention. The both-feature and weak-only models share initialization, labels, samples, weak coordinates, and noise; only the strong input channel is set to zero in the weak-only condition. We track common feature-response functionals in both trajectories. This setup separates five statements:

1. *relative transfer*: the strong cue initially increases the weak response rate, so $d(\tau) > 0$;
2. *rate suppression*: the both-feature weak response later grows more slowly than its counterfactual, so $d(\tau) < 0$;
3. *outcome suppression*: the both-feature weak response falls below its matched counterfactual, so $\Delta(\tau) < 0$;
4. *weak-only nonlearnability under the specified gate*: the weak-only path does not first hit the preregistered target by the horizon; and
5. *causal starvation*: outcome suppression occurs while the weak-only counterfactual demonstrably learns under that first-hit gate.

A failed weak-only gate therefore licenses neither a causal-starvation certificate nor evidence that starvation is absent; causal status is indeterminate under the specified target and horizon. A transfer-to-starvation trajectory is a particular causally certified path with an initial positive excursion before outcome suppression, not a default description of weak learning failure. This distinction is empirical as well as logical: at the fixed nonlinear cell, tanh has complete transfer-to-starvation certificates in 3/8 observed seeds, whereas GRU has drift and response crossings in 5/8 but fails the causal learnability gate in every seed. These counts do not rank architectures or estimate prevalence.

The diagnostic has an exact finite-width dynamical basis. For signed sample logits r , full-batch logistic gradient flow gives $\dot{M}_w = c_w^\top q(r)/n$, with $q_i(r) = \sigma(-r_i)$ the cross-entropy weighting and $c_w = J\nabla M_w$ the response-specific response-logit geometry. No DMFT is needed for this chain-rule identity. For the paired gap, the pathwise balance $\Delta = P - N$ allows arbitrary drift sign changes: outcome suppression occurs only when accumulated suppression overtakes accumulated help, and causal starvation still requires the independent weak-only gate. Prediction is partial. The realized-initialization value $d(0)$ and, where proved, local jet information such as $d'(0)$ give rigorous local or assumption-scoped sufficient signs. Stronger global parameter-level prediction from ρ , lag, and RNN geometry remains unresolved because nonvacuous positive-lag product-ball and kernel-movement constants are unavailable. CDC then uses the same matched weak-only information as a local mechanism-level control, not as a claimed general remedy.

Contributions.

1. We operationalize a paired causal test that separates relative transfer, rate suppression, outcome suppression, weak-only gate failure, and causal starvation through direct drift, first-hit, AUC, and finite-step certificates. A post-hoc terminal β profile reports threshold sensitivity from existing outcomes without replacing the preregistered first-hit gate.
2. We report the full findings sequence rather than only positive endpoints. Controlled E1 has positive paired weak-AUC deficits and learnable weak-only controls in all measured cells at lags 0, 2, and 4, while lag 8 fails the gate and is causally indeterminate. E-NL separates 3/8 tanh transfer-to-starvation certificates from 5/8 ungated GRU drift/response crossings. E19 remains a failed five-endpoint pilot; E20 remains the submission-primary empirical claim and supports only fresh restricted drift/response event labels. E26 is negative/indeterminate after all records fail their specified gate; E27 is negative without an architecture ranking; and E28 yields a comparator-restricted mixed CDC tradeoff. Archives distinguish historically unreconstructible source from clean, reconstructible source and from self-contained replay packages.
3. We derive exact finite-width logistic response flow and full signed-logit empirical-NTK/response dynamics. The weak response obeys $\dot{M}_w = c_w^\top q(r)/n$, explicitly separating cross-entropy weighting from response-specific geometry. A closed projected equation $\dot{M} = Gg$ requires exact parameter-independent logit realization; the initialization-frozen nonlinear logistic surrogate is exact internally, while comparison with a trained nonlinear model is conditional on supplied kernel-movement bounds.
4. We prove the pathwise cumulative drift balance for arbitrary sign changes, with the exact single-crossing tail-area criterion as a corollary. We add a conditional noiseless rank-one rate-crossing criterion and an exact realized- initialization dense-linear $d(0)$ threshold with its existential strict local outcome-sign implication. Independently proved product-ball bounds then imply unique crossing, outcome suppression, nondegenerate weak-only learnability, or finite-horizon safe transfer; no current component constructs these bounds for general positive-lag RNNs. We also prove stability of quantitatively isolated transverse first hits under uniform-error, radius, and mesh hypotheses, and an exact zero-lag recurrence-invisible six-scalar wide limit for arbitrary finite realized recurrent gain, with immediate deterministic suppression and weak-only learnability under the noiseless cue law.
5. As local mechanism-level control, in the fixed implemented Euclidean/Frobenius chart, feasible uncapped target-attaining CDC is the unique minimum-Euclidean-norm correction satisfying zero instantaneous strong-drift change and the weak-drift lower bound. Its finite-step guarantee is local and one-step. The ablations establish neither SOTA performance, canonical parity, broad practical superiority, nor best final retention.

Claim boundary. We do not claim that recurrent networks universally undergo transfer-to-starvation, that a general joint CE-RNN optimization-time DMFT has been solved, or that an analytic ρ_c or universal (ρ, lag) phase boundary is known. A failed weak-only gate is causal indeterminacy under the specified target and horizon: it establishes neither causal starvation nor its absence, and it does not show that a dataset is intrinsically unlearnable. The exact initialization threshold and proved local jets provide local or assumption-scoped signs, not a general predictor; at positive lag, the threshold is realization-dependent and its trajectory implications require uniform product-ball constants that are not currently constructed. The frozen empirical-NTK surrogate is exact for its initialization-linearized tangent model, not for trained tanh/GRU networks: no required instantaneous or secant-kernel stability bound is proved here. E19 is a failed broad pilot. E20 validates only binary drift/response-crossing classification at one width/task cell; it does not validate phase, causal starvation, weak-only learnability, calibrated trajectories, crossing times, magnitudes, or prevalence. E20 remains submission-primary with only its frozen drift/response-crossing endpoints primary; E26–E28 do not outcome-dependently reprioritize it. The eight-seed nonlinear proportions are descriptive, not population prevalence. E26 uses real benchmark pixels with a generated removable cue and therefore supplies semi-real, not natural-spurious-correlation, evidence; its absent weak-only first hits make zero causal certificates indeterminate. E27 supports no beta-robust architecture claim or architecture ranking. E28 supports a joint tradeoff only against one of two non-canonical comparators; CDC does not beat the non-canonical PCGrad-style shadow-target rescue under the frozen joint criterion. No SOTA, broad CDC superiority, or canonical parity is claimed. The general DMFT solver remains deliberately unimplemented where proof obligations are unmet.

2 RELATED WORK

Gradient starvation. Pezeshki et al. (2021) identify feature starvation under cross-entropy and motivate Spectral Decoupling. Our focus is a recurrent paired causal diagnosis: the same-initialization weak-only intervention tests the learnability premise, exact finite-width dynamics separate cross-entropy weighting from response-specific geometry, and pathwise balance separates rate suppression from outcome suppression and causally gated starvation. A failed weak-only gate is indeterminate rather than evidence for either starvation or its absence.

Shortcut learning and spurious-correlation robustness. Shortcut learning and simplicity bias explain why predictive but unintended cues can dominate learned representations (Geirhos et al., 2020; Shah et al., 2020). Group DRO, Just Train Twice, and last-layer retraining instead improve worst-group performance under spurious correlations (Sagawa et al., 2020; Liu et al., 2021; Kirichenko et al., 2023). These phenomena motivate our setting, but they do not supply the paired response-level causal criterion used here. We are not proposing a competing robustness algorithm: our diagnostic requires a stronger matched-shadow information setting. E26 adds a generated-cue intervention on real MNIST and FashionMNIST core pixels, but it is not an unmodified natural-spurious-correlation benchmark and supports no such claim.

Learning dynamics in recurrent and wide networks. Neural tangent kernels describe infinite-width gradient dynamics (Jacot et al., 2018), and wide-network linearization motivates frozen-tangent comparisons (Lee et al., 2019). Our initialization-frozen logistic surrogate retains the nonlinear loss but is exact only for its tangent model. Its comparison with a trained finite-width network is conditional on explicit instantaneous or secant-kernel movement bounds, which we do not prove for tanh or GRU. Ger and Barak (2026) derive low-dimensional overlap learning dynamics for low-rank RNNs and show that overlaps invisible to the loss can retain training history. This directly cautions against closing a recurrent learning theory using only visible response coordinates. Clark et al. (2026) develop a mean-field account of structure and disorder in task-trained recurrent circuits in a long-time Langevin/Gibbs setting. That equilibrium construction is distinct from the paired, noiseless optimization-time gradient flow studied here. Accordingly, we claim neither the first learning theory for RNNs nor a completed general RNN DMFT; our contribution is the matched causal separation supported by exact finite-width response and pathwise attribution results.¹

¹The descriptions of the 2026 sources were paraphrased from the linked public sources. Content was rephrased for compliance with licensing restrictions.

Gradient surgery. PCGrad projects conflicting task gradients (Yu et al., 2020); Bloop stabilizes an auxiliary projection using an exponential moving average (Hsieh et al., 2024). Projection itself is not our novelty. CDC protects the gradient of a theory-defined strong-feature response and takes its target from a matched causal weak-only shadow. This stronger information setting must be reported: CDC requires a constructible ablation, a second model, lockstep training, and full-batch gradients. In E28 the evaluated Bloop-style shadow-target rescue and PCGrad-style shadow-target rescue are non-canonical (`canonical: false`); the frozen joint weak-rescue/strong-preservation tradeoff holds only against the former, and CDC does not beat the latter under that criterion. The study establishes neither SOTA, canonical parity, nor broad CDC superiority.

3 PAIRED CAUSAL DIAGNOSIS

Weak learning failure alone is not gradient starvation. For condition $C \in \{B, W\}$, the both-feature and weak-only models share initialization, labels, samples, weak coordinates, and noise; only the strong input channel is ablated under W. The matched weak-only path tests whether the weak signal can meet the specified target without the strong cue; it is an identification premise rather than a second observation of the same failure. Let $M_w^C(\tau)$ be the common weak-feature response, and define

$$\Delta(\tau) = M_w^B(\tau) - M_w^W(\tau), \quad d(\tau) = \Delta'(\tau), \quad \Delta(0) = 0. \quad (1)$$

This pairing separates claims that are often conflated: $d > 0$ is *relative transfer*; $d < 0$ is *rate suppression*; $\Delta < 0$ is *outcome suppression*; failure to first hit a preregistered target β by horizon H is *weak-only nonlearnability under that gate*; and *causal starvation* requires outcome suppression together with a successful, nondegenerate weak-only first hit. A response crossing without weak-only learnability is descriptive, not a causal certificate. Conversely, the failed gate is not evidence that starvation is absent; causal status is indeterminate under the specified target and horizon.

For signed logits $r_i^C = y_i f_C(x_i; \theta_C)$ and full-batch logistic gradient flow, the exact finite-width chain rule is

$$\dot{M}_a^C = \frac{1}{n_C} \sum_i \langle \nabla M_a^C, \nabla r_i^C \rangle \sigma(-r_i^C). \quad (2)$$

We state this as a proposition rather than a theorem: the derivation is elementary; the contribution is its exact finite-width scope and the causal separation it enables. With sample Jacobian J , empirical NTK $\Theta = JJ^\top$, response cross-kernel $c_a = J \nabla M_a$, and cross-entropy weights $q_i(r) = \sigma(-r_i)$, the exact dynamics are $\dot{r} = \Theta q(r)/n$ and $\dot{M}_a = c_a^\top q(r)/n$. Thus weak-response motion separates sample weighting $q(r)$ from response-specific geometry c_w ; no DMFT or response-coordinate closure is needed for this identity. Freezing (Θ, c_a) at initialization defines a nonlinear logistic tangent surrogate. Its comparison with nonlinear training is conditional on supplied kernel-movement bounds; event agreement is not a substitute for that premise.

4 PATHWISE SEPARATION OF HELP, SUPPRESSION, AND CAUSAL STATUS

For any absolutely continuous Δ , let $P(t) = \int_0^t [d(u)]_+ du$ and $N(t) = \int_0^t [-d(u)]_+ du$. Our general bookkeeping proposition gives the exact identity

$$\Delta(t) = P(t) - N(t), \quad \Delta(t) < 0 \iff N(t) > P(t). \quad (3)$$

It allows arbitrary sign changes and immediate suppression: outcome suppression occurs only when accumulated suppression N overtakes accumulated help P . Under one $+\rightarrow-$ drift crossing at τ_d , the identity reduces to the tail-area criterion, in which negative post-crossing area must exceed the positive peak area. Neither rate suppression nor outcome suppression establishes causal starvation without the independent weak-only learnability gate; a failed gate leaves causal status indeterminate. The appendix proves this statement, a conditional rank-one sufficient criterion for a unique rate crossing, and stability of quantitatively isolated transverse first hits under uniform-error, radius, and mesh hypotheses.

The appendix also gives an initialization-conditioned pre-trajectory result. In the noiseless positive rank-one dense-linear pair, set $K = G_{ws}(0)$, $A = G_{ww}(0)$, $s_B = \sigma[-(\rho M_s(0) + M_w(0))]$, and

$s_W = \sigma[-M_w(0)]$. The exact initial rate gap is

$$d_0 = s_B(\rho K + A) - s_W A, \quad d_0 > 0 \iff K > \frac{A}{\rho} \left(\frac{s_W}{s_B} - 1 \right). \quad (4)$$

This is partial, initialization-conditioned prediction: it uses the realized d_0 , not a trained trajectory or a universal (ρ, lag) boundary. Shared initialization has $\Delta(0) = 0$, so $d_0 < 0$ and $d_0 > 0$ respectively imply strict local outcome suppression and transfer for all sufficiently small positive times. This derivative argument is existential and gives no quantitative horizon; the executable certificate reports values within its caller tolerance as boundary/undetermined. Under independently supplied uniform product-ball bounds $-\Lambda \leq d' \leq -\lambda < 0$ up to $h_R = R/S_R$, one obtains $d_0/\Lambda \leq \tau_d \leq d_0/\lambda$, and $h_R > 2d_0/\lambda$ implies outcome suppression. A separate positive weak-only drift bound gives causal learnability only for a target strictly above the initial weak response; a target already met at initialization is degenerate and cannot gate causal starvation. Alternatively, $|d'| \leq J$ and $d_0 - Jh_R > 0$ exclude suppression on that finite horizon. These are local or assumption-scoped sufficient signs, not a general parameter-level predictor. No current implementation constructs nonvacuous general positive-lag constants from ρ , lag, and recurrent geometry.

A proved local-jet example occurs in the deterministic finite-horizon wide limit at zero lag, zero background input, and arbitrary finite realized recurrent gain: all pre-final states vanish, so W is loss-invisible. For the explicitly noiseless (`cue_noise` = 0) positive rank-one law, its symmetric reference has $\bar{d}(0) = 0$ and $\bar{d}'(0) = -\rho^2/2 < 0$, so suppression starts immediately rather than after transfer; the weak-only response reaches every finite positive target. This is a recurrence-invisible special case, not a predictor for general positive lag. A general positive-lag optimization-time DMFT remains open.

As local mechanism-level control, Counterfactual Drift Correction (CDC) uses the weak-only shadow as a local target. In the fixed implemented Euclidean/Frobenius parameter chart, let $a = \nabla m_s$, $p = \nabla m_w$, $v = -\nabla L$, and let q be the projection of p onto $a^\perp$. For positive deficit $\delta = [F_w^W - p \cdot v]_+$, uncapped CDC takes $v' = v + (\delta/\|q\|^2)q$ when feasible. In that feasible uncapped and target-attaining case, it is the unique minimum-Euclidean-norm correction satisfying zero instantaneous strong-drift change and the weak-drift lower bound. Its finite-step strong-response guarantee is local and one-step, of order $O(\eta^2)$ under local smoothness. These are fixed-chart, full-batch statements, not trajectory, SOTA, canonical-parity, or practical-superiority claims.

5 EXPERIMENTS AND AUDIT PROTOCOL

The evidence combines controlled synthetic studies with one semi-real generated-cue intervention. E1 uses a dense linear RNN with four strengths and seed IDs 4–11: each lag therefore has 32 ρ -seed cells but only eight independent seed IDs. E-NL evaluates one $(\rho, \text{lag}) = (4, 2)$ cell with eight seeds each for tanh and GRU, paired lockstep training, direct-autograd response drift, horizon H , and preregistered first-hit target $\beta = 0.5$. Historical E3 compares CDC with ERM and seven mitigation arms; the matched-shadow methods use stronger information and roughly twice the model work.

The externally sealed clean-source extensions were executed at commit 9ebbc617a9aaad5c3c19e9d5fd4fa08f2d3517d0. E26 crosses four data seeds with eight model seeds on each of MNIST 3-vs-8 and FashionMNIST 0-vs-6, retaining real core pixels and ablating only an exactly removable generated cue. E27 contains 192 nonlinear records over three fixed cells, tanh/GRU, and prospective $\beta \in \{0.25, 0.5, 1, 2\}$. E28 contains 256 method-records for ERM, CDC, Bloop-style shadow-target rescue, and PCGrad-style shadow-target rescue. The two named comparators are non-canonical (`canonical`: false); no canonical parity claim is made.

The frozen-tangent sequence is intentionally fail-closed. E19 sealed predictions for five endpoints before training on 16 architecture-seed records. After that broad rule failed, E20 froze only drift- and response-crossing labels as primary on a fresh 4×8 crossed data/model-seed factorial per architecture. E20 remains the submission-primary empirical study. Its 10,000-replicate two-way bootstrap resamples the two observed seed axes. Surviving manifests, predictions, scores, hashes, and seals make E19/E20 auditable, but their exact historical dirty source snapshots were never archived; they are not replayable, and regeneration is a new study.

Table 1: Final nonlinear and frozen-tangent audits. Counts describe the observed synthetic records, not population prevalence. E20 confidence intervals resample both seed axes.

Study	Endpoint	Overall	tanh	GRU
E-NL	causal certificate	3/16	3/8	0/8
E-NL	response crossing	8/16	3/8	5/8
E19	phase / drift / response	12/16 / 14/16 / 14/16	—	—
E19	causal / learnability	14/16 / 12/16	—	—
E20	drift event	60/64, CI [.8125, 1]	32/32	28/32
E20	response event	64/64, CI [1, 1]	32/32	32/32

Table 2: Clean-source E26–E28 completed outcomes. Brackets are 95% CIs; E27 Holm-adjusted $p = 1$ for each architecture.

Study	Endpoint / contrast	Estimate [95% CI]
E26	MNIST weak AUC gap	−0.0012362842136667493 [−0.0034405428460004646, 0.0010150591371180453]
E26	FashionMNIST weak AUC gap	−0.00010921728159018996 [−0.004129384520886106, 0.003182670049955049]
E27	tanh / GRU all- β	0.2916666666666667 [0.16666666666666666, 0.4166666666666667] / 0 [0, 0]
E28	CDC–Bloop-style shadow-target rescue weak rescue	0.0021982593992110656 [0.00009779187294930126, 0.003277366267916477]
E28	Bloop-style shadow-target rescue–CDC trajectory / final deviation	3.4651361294978416 [3.3443827215131754, 3.5849715262780952] / 0.6422362388111651 [0.6288088704226539, 0.6563168084365315]
E28	CDC–PCGrad-style shadow-target rescue weak rescue	−0.00015351228407674823 [−0.00047905548592828785, 0.00013248012418980214]
E28	PCGrad-style shadow-target rescue–CDC trajectory / final deviation	−0.48709889128076556 [−0.689737894428417, −0.27379327373499074] / −0.2565436437726021 [−0.3016841153614223, −0.2087651835754514]

6 RESULTS

Paired deficits and causal diagnoses. At lags 0, 2, and 4, E1 has a positive paired weak-AUC deficit in all 32 ρ -seed cells per lag, and all corresponding weak-only controls pass the specified learnability gate; mean deficits are 5.76, 8.93, and 6.11. At every fixed lag, the mean deficit increases over the measured ρ grid. All 32 lag-8 cells fail the weak-only gate and are weak-only nonlearnable and causally indeterminate under $\beta = 0.25$, $H = 5$; they are neither causal-starvation certificates nor evidence of absence. At E-NL’s fixed $(\rho, \text{lag}) = (4, 2)$ cell, tanh has complete transfer-to-starvation certificates in 3/8 observed seeds. GRU has drift and response crossings in 5/8 but 0/8 causal certificates because every weak-only path fails $\beta = 0.5$. Thus rate/outcome suppression differs from causal starvation. These observed counts support neither an architecture ranking nor a prevalence claim.

Gate-indeterminate breadth and comparator-restricted control. E26 is negative/indeterminate: MNIST 3-vs-8 and FashionMNIST 0-vs-6 each have 0/32 weak-only-learnable and 0/32 causal records. The failed gate rejects a false- positive causal-starvation interpretation, but it also means that zero certificates cannot establish absence of starvation or intrinsic dataset unlearnability. E27’s all- β architecture endpoint is false for both architectures and was not used for ranking. Table 2 reports the exact macro estimates. For E28, weak rescue is $-|\text{weak AUC gap}|$ (higher is better), so “CDC–comparator” positive favors CDC; trajectory and final deviation are absolute deviations from paired ERM (smaller is better), so “comparator–CDC” positive favors CDC. The frozen joint tradeoff is true only versus the non-canonical Bloop-style shadow-target rescue. It is false versus the non-canonical PCGrad-style shadow-target rescue, so CDC does not beat that variant under the frozen joint criterion. No SOTA, canonical parity, or broad CDC superiority follows.

β sensitivity without retraining. A post-hoc audit computes each completed weak-only terminal response $B_{W,s}(H)$ and the fraction above candidate thresholds. Tanh terminal minimum, mean, and maximum are 4.41313266754, 4.49664855003, and 4.67605876923 (8/8 above 0.5); the corresponding GRU values are 0.0828229486942, 0.129690139554, and 0.187142759562 (0/8). Thus GRU is not barely below the selected operating point. This terminal profile is not generally equivalent to first-hit learnability on a nonmonotone path and supports neither a new confirmatory endpoint nor an architecture ranking.

E19 falsification and E20 restricted event-label result. E19 obtained phase/drift/response/causal/learnability correctness 12/16, 14/16, 14/16, 14/16, 12/16, but failed because tanh learnability was $4/8 < 5/8$. Its tanh response-gap, weak-response, and drift RMSE were 2.367, 2.394, and 0.486; the corresponding GRU values were 0.0163, 0.0281, and 0.0107. E20 later passed only the restricted event rule. Its secondary phase/causal/learnability correctness was 52/64, 56/64, 49/64 (tanh learnability 17/32); drift/response crossing-time bias was

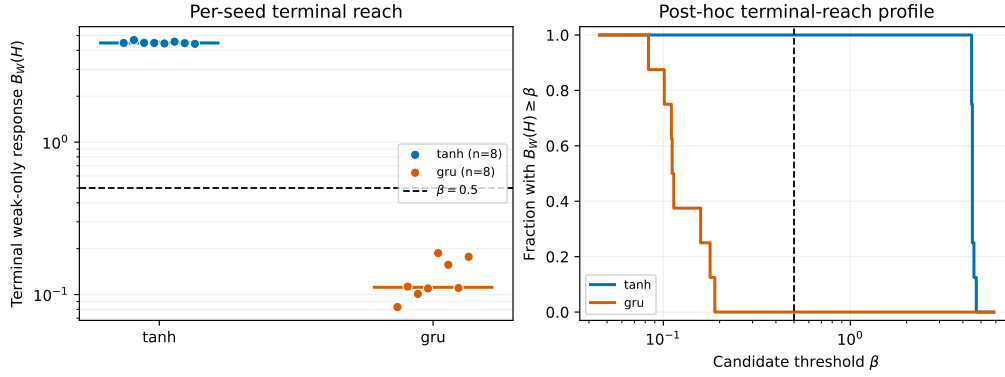


Figure 1: Post-hoc weak-only terminal-response audit from the completed E-NL run. Points show all eight seeds, horizontal segments show medians, and the dashed line marks the preregistered $\beta = 0.5$. The right panel sweeps candidate thresholds without rerunning training.

$-0.822 / -1.189$; and tanh response-gap, weak-response, and drift RMSE were 2.455, 2.489, and 0.490. Hence E20 is an uncalibrated one-cell event classifier, not phase, causal, learnability, time, magnitude, prevalence, or kernel-stability validation.

Historical E3 local mechanism-control ablation. In historical E3, all five shadow-based mitigation arms reduce the dense weak AUC gap from 10.5155 to approximately zero. CDC alone preserves instantaneous strong drift to 4.77×10^{-7} , but it does not have the best final strong response. This supports the fixed-chart local mechanism-level guarantee in that historical synthetic setting, not SOTA performance or broad practical superiority; E28 separately passes the frozen joint tradeoff only against the non-canonical Bloop- style shadow-target rescue and not against the non-canonical PCGrad-style shadow- target rescue.

7 LIMITATIONS AND CONCLUSION

Weak learning failure alone is not gradient starvation. The matched intervention first asks whether the weak-only path learns under the specified target and horizon; a failed gate makes causal status indeterminate rather than proving either starvation or its absence. Conditional on that premise, rate suppression becomes outcome suppression only when accumulated negative relative drift overtakes accumulated positive drift. The paper’s strongest chain is therefore exact finite- width dynamics $\rightarrow$ matched causal intervention $\rightarrow$ pathwise attribution $\rightarrow$ local mechanism-level control. Proposition 1 and the cumulative balance are deliberately elementary; their value is the exact finite- width scope and the separation of weak learning failure, rate, outcome, and causal claims. Realized $d(0)$ and proved local jets give local or assumption-scoped signs, while general positive-lag prediction from ρ , lag, and recurrent geometry remains open.

The empirical ceiling is explicit: controlled synthetic task families plus one semi-real generated-cue intervention, limited fixed cells and seed grids, and no estimate of universal event prevalence. E19 remains failed, and E20 remains the submission-primary empirical study with only its restricted frozen drift/response- crossing classification endpoints primary. E26 is not a natural shortcut benchmark, and $0/32$ weak-only learnability per dataset makes its zero causal count indeterminate rather than evidence of no starvation or intrinsic dataset unlearnability. E27’s beta-robust endpoint is negative for both architectures and supports no ranking. E28’s comparators are non-canonical; only the Bloop-style shadow-target rescue satisfies the frozen joint tradeoff, while CDC does not beat the PCGrad-style shadow-target rescue under that criterion, so neither SOTA, canonical parity, nor broad CDC superiority follows. The unrun Waterbirds adapter remains a minibatch head-coordinate surrogate outside the full-batch matched- shadow theorem, and its intercept is not an identified bird-shape response; we make no Waterbirds or natural-spurious-correlation claim. E19/E20 are integrity- auditable but historically unreplayable, whereas E26–E28 used clean reconstructible source at commit `9ebbc617a9aaad5c3c19e9d5fd4fa08f2d3517d0`. Their compact archives omit

192 and 1,792 records/** files, respectively, and therefore are auditable but not standalone replay packages.

REPRODUCIBILITY STATEMENT

The appendix states all theorem assumptions and proofs; gives synthetic and semi-real data, cue-generation, optimization, seed, censoring, and inference protocols; maps claims to artifacts and executable guards; and reports failed, negative, indeterminate, mixed, and successful studies. Resolved configurations, environment locks, selection and initialization manifests, authorization seals, tracked figures, compact outcomes, a dependency-free artifact verifier, and the full test suite are provided. The supplement distinguishes reconstructible clean-source E26–E28, auditable but historically unreplayable E19/E20, and compact archives that omit record bytes and thus are not standalone replay packages.

AI USE STATEMENT

Generative AI tools were used to assist with code drafting and refactoring, manuscript editing and organization, and review-item tracking. They were not used to generate or infer experimental outcomes: every reported number was checked against saved artifacts or frozen evaluation records. The authors reviewed the AI-assisted text and code, ran the test and artifact-validation suites, checked claim scope against the theorem and evidence ledgers, and take responsibility for the final content, including all AI-assisted text, claims, code, and artifacts.

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A PAIRED CAUSAL DIAGNOSIS AND DEFINITIONS

A.1 PAIRED LOGISTIC GRADIENT FLOW

For condition $C \in \{B, W\}$, let $\theta_C(\tau) \in \mathbb{R}^P$ follow full-batch gradient flow on signed logits $r_i^C(\theta) = y_i f_C(x_i; \theta)$:

$$L_C(\theta) = \frac{1}{n_C} \sum_{i=1}^{n_C} \log(1 + e^{-r_i^C(\theta)}), \quad \dot{\theta}_C = -\nabla L_C(\theta_C). \quad (5)$$

The two conditions share initialization and all data except that the strong channel is zeroed under W. This matched weak-only path tests whether the weak signal can meet the specified target without the strong cue; it is the causal identification premise that distinguishes suppression from an unsupported weak- learning label. Let M_s^C, M_w^C be common differentiable strong and weak response functionals. Shared initialization gives

$$\Delta(0) = 0, \quad \Delta(\tau) = M_w^B(\tau) - M_w^W(\tau). \quad (6)$$

Optimization time τ is distinct from sequence index k .

The controlled dense linear RNN is

$$h_{k+1} = Wh_k + Bx_k, \quad f(x) = c^\top h_T, \quad (7)$$

with $W_{ij}(0) \sim \mathcal{N}(0, g^2/N)$, $B_{ia}(0) \sim \mathcal{N}(0, 1/N)$, and $c_i(0) \sim \mathcal{N}(0, 1/N)$. The width scaling of B is essential for a non-divergent projected metric. We additionally study tanh RNNs and GRUs. For nonlinear models the two responses are common odd symmetric unit-probe contrasts, $[f(x_a^+) - f(x_a^-)]/2$, not assumed additive coordinates of every data logit.

A.2 WEAK LEARNING FAILURE, CAUSAL LABELS, AND METRICS

Let

$$d(\tau) = \Delta'(\tau) = F_w^B(\tau) - F_w^W(\tau). \quad (8)$$

We call $d > 0$ relative transfer and $d < 0$ relative rate suppression. We call $\Delta < 0$ *outcome suppression*, whether it follows a positive excursion or is present from the start of optimization. A response crossing is the special case in which Δ returns to zero after a positive excursion and then becomes negative. Weak-only nonlearnability under the specified gate means that the weak-only trajectory does not first hit the preregistered target β within horizon H . We reserve *causal starvation* for outcome suppression when that weak-only trajectory does reach the target at a nondegenerate positive time. Thus weak learning failure alone is not gradient starvation. If the weak-only model does not reach β , an outcome crossing is descriptive but not a causal- starvation certificate; the failed

gate also is not evidence that starvation is absent. Causal status is indeterminate under the specified target and horizon.

We report first-hitting-time delay when both hits are observed. Formally, $\tau_W(\beta; H) = \inf\{\tau \in [0, H] : M_w^W(\tau) \geq \beta\}$, with $\inf \emptyset = \infty$; the learnability gate requires $\tau_W(\beta; H) \in (0, H]$, excluding targets met at initialization. The horizon-wide weak AUC gap is

$$\mathcal{A}_w = \int_0^H (M_w^W(\tau) - M_w^B(\tau)) d\tau, \quad (9)$$

computed by the logged-grid trapezoid rule. To expose rather than hide sensitivity to the single preregistered β , Section H also reports the threshold-free terminal response $B_W(H) = M_w^W(H)$ across seeds and its post-hoc threshold profile. Terminal thresholding is not generally equivalent to the preregistered first-hit event on a nonmonotone trajectory, so the profile is a sensitivity diagnostic rather than a replacement gate. GSI-5 remains a margin-weight diagnostic, not a definition of starvation.

B EXACT FINITE-WIDTH DYNAMICS FOR THE PAIRED DIAGNOSIS

The mechanism-level decomposition is exact before any width limit: with $q_i(r) = \sigma(-r_i)$ and response-specific geometry $c_w = J\nabla M_w$, the weak response obeys $\dot{M}_w = c_w^\top q(r)/n$. The vector $q(r)$ records cross-entropy weighting across signed sample logits, while c_w records how the chosen weak response couples to those logits through parameter space. No DMFT or two-response closure is needed for this identity; such closures arise only in stronger predictive approximations.

Proposition 1 (Universal response-logit cross-kernel flow). *At every finite width and every optimization time at which M_a^C is differentiable,*

$$\dot{M}_a^C = \frac{1}{n_C} \sum_i K_{ai}^C \sigma(-r_i^C), \quad K_{ai}^C = \langle \nabla M_a^C, \nabla r_i^C \rangle. \quad (10)$$

No width limit or two-mode realization is required. We label this elementary chain-rule identity as a proposition: its importance here is the exact finite-width scope and the causal separation it enables, not proof difficulty.

Corollary 2 (Closed projected flow under exact realization). *If the signed logits admit the exact parameter-independent representation $r_i^C = \sum_b z_{ib}^C M_b^C$, then*

$$\dot{M}_a^C = \sum_b G_{ab}^C g_b^C, \quad G_{ab}^C = \langle \nabla M_a^C, \nabla M_b^C \rangle, \quad g_b^C = \frac{1}{n_C} \sum_i z_{ib}^C \sigma(-z_i^C \cdot M^C). \quad (11)$$

If $r_i^C = z_i^C \cdot M^C + r_i^\perp$, the exact equation instead contains

$$R_a^C = \frac{1}{n_C} \sum_i \sigma(-r_i^C) \langle \nabla M_a^C, \nabla r_i^\perp \rangle, \quad \dot{M}_a^C = (G^C g^C)_a + R_a^C. \quad (12)$$

Corollary 3 (Exact full empirical-NTK dynamics). *Let $J^C(\tau) \in \mathbb{R}^{n_C \times P}$ have row $\nabla r_i^C(\theta_C(\tau))^\top$, and define*

$$\Theta^C(\tau) = J^C(\tau) J^C(\tau)^\top, \quad c_a^C(\tau) = J^C(\tau) \nabla M_a^C(\theta_C(\tau)). \quad (13)$$

Writing $q(r) = \sigma(-r)$ componentwise, the exact finite-width flow is

$$\dot{r}^C = \frac{1}{n_C} \Theta^C(\tau) q(r^C), \quad \dot{M}_a^C = \frac{1}{n_C} c_a^C(\tau)^\top q(r^C). \quad (14)$$

No width limit or response-coordinate realization is required.

Definition 4 (Initialization-frozen nonlinear logistic surrogate). For one condition, let $\Theta_0 = \Theta(0)$ and $c_0 = c_a(0)$. Define

$$\dot{\bar{r}} = \frac{1}{n} \Theta_0 q(\bar{r}), \quad \dot{\bar{M}} = \frac{1}{n} c_0^\top q(\bar{r}), \quad (\bar{r}(0), \bar{M}(0)) = (r(0), M(0)). \quad (15)$$

The gate remains nonlinear. This system is exact for the initialization-linearized tangent model, not for the trained nonlinear network.

Theorem 5 (Conditional continuous and secant-SGD comparison). *Use Euclidean and induced operator norms. Suppose on $[0, t]$ that $\|\Theta(u) - \Theta_0\| \leq B_\Theta$, $\|c(u) - c_0\| \leq B_c$, and both logistic weight vectors have norm at most V . Set $a = \|\Theta_0\|/(4n)$,*

$$\phi_a(t) = \frac{e^{at} - 1}{a}, \quad \psi_a(t) = \frac{e^{at} - 1 - at}{a^2}, \quad (16)$$

with $\phi_0(t) = t$ and $\psi_0(t) = t^2/2$. Then

$$\|r(t) - \bar{r}(t)\| \leq \frac{B_\Theta V}{n} \phi_a(t), \quad (17)$$

$$|M(t) - \bar{M}(t)| \leq \frac{B_c V}{n} t + \frac{\|c_0\| B_\Theta V}{4n^2} \psi_a(t). \quad (18)$$

For explicit full-batch SGD, let $\delta\theta_k = \theta_{k+1} - \theta_k$ and define exact segment kernels

$$\tilde{J}_k = \int_0^1 J(\theta_k + s\delta\theta_k) ds, \quad \tilde{\Theta}_k = \tilde{J}_k J(\theta_k)^\top, \quad (19)$$

$$\tilde{m}_k = \int_0^1 \nabla M(\theta_k + s\delta\theta_k) ds, \quad \tilde{c}_k = J(\theta_k) \tilde{m}_k. \quad (20)$$

The true increments are exactly $r_{k+1} - r_k = \eta \tilde{\Theta}_k q(r_k)/n$ and $M_{k+1} - M_k = \eta \tilde{c}_k^\top q(r_k)/n$. If $\eta > 0$ and, for every $0 \leq j < k$, $\|\tilde{\Theta}_j - \Theta_0\| \leq D_\Theta$, $\|\tilde{c}_j - c_0\| \leq D_c$, and $\|q(r_j)\|, \|q(\bar{r}_j)\| \leq V$, then the explicit-Euler version of Eq. (15) obeys

$$\|r_k - \bar{r}_k\| \leq \frac{D_\Theta V}{n} \Phi_k, \quad (21)$$

$$|M_k - \bar{M}_k| \leq \frac{k\eta D_c V}{n} + \frac{\eta \|c_0\| D_\Theta V}{4n^2} \Psi_k, \quad (22)$$

where

$$\Phi_k = \frac{(1 + \eta a)^k - 1}{a}, \quad \Psi_k = \frac{(1 + \eta a)^k - 1}{\eta a^2} - \frac{k}{a}, \quad (23)$$

with limits $\Phi_k = k\eta$ and $\Psi_k = \eta k(k-1)/2$ at $a = 0$. Apply the bounds separately to B, W and add them to bound the paired response gap. Frozen signs or brackets transfer only when their strict margins exceed the corresponding summed errors; first hitting times additionally require Corollary 14's prehistory, transversality, radius, and mesh hypotheses.

Theorem 5 is conditional: this paper does not prove the required instantaneous or secant-kernel movement bounds for tanh or GRU. A held-out event-classification result cannot substitute for that premise. In particular, the theorem is finite-width and is not a DMFT closure.

For channel-localized modes of the dense linear RNN, differentiation across the disjoint (c, B, W) blocks gives the exact finite-width geometry

$$G_{ab} = p_a^\top p_b + \mathbf{1}\{\text{ch}(a) = \text{ch}(b)\} q_a^\top q_b + \langle S_a, S_b \rangle_F, \quad (24)$$

where $p_a = W^{\ell_a} b_a$, $q_a = (W^{\ell_a})^\top c$, and

$$S_a = \sum_{j=0}^{\ell_a-1} (W^j)^\top c (W^{\ell_a-1-j} b_a)^\top. \quad (25)$$

Theorem 6 (Exact realized-initialization rate-gap criterion). *In the noiseless positive rank-one dense-linear pair, let the strong cue have strength $\rho > 0$ at lag zero, the unit weak cue have lag ℓ , and define*

$$K = G_{ws}(0) = b_s^\top W^\ell b_w, \quad (26)$$

$$A = G_{ww}(0) = \|W^\ell b_w\|^2 + \|(W^\ell)^\top c\|^2 + \|S_w\|_F^2, \quad (27)$$

$$S_w = \sum_{j=0}^{\ell-1} (W^j)^\top c (W^{\ell-1-j} b_w)^\top, \quad (28)$$

$$s_B = \sigma[-(\rho M_s(0) + M_w(0))], \quad s_W = \sigma[-M_w(0)]. \quad (29)$$

(The recurrent sum is zero for $\ell = 0$.) The exact initial equal-time weak-rate gap is

$$d_0 = s_B(\rho K + A) - s_W A. \quad (30)$$

Consequently, initial transfer, suppression, and the boundary occur respectively when

$$K \geq \frac{A}{\rho} \left(\frac{s_W}{s_B} - 1 \right) \quad \text{or equality holds.} \quad (31)$$

This is an exact pre-trajectory criterion from the realized initialization. Since shared initialization gives $\Delta(0) = 0$ and $\Delta'(0) = d_0$, it also implies

$$d_0 < 0 \implies \Delta(t) < 0, \quad d_0 > 0 \implies \Delta(t) > 0 \quad \text{for all sufficiently small } t > 0. \quad (32)$$

These strict local outcome signs are existential and provide no quantitative horizon. A numerical d_0 within the executable certificate's caller tolerance is reported as boundary/undetermined and licenses neither sign. The criterion is not a distribution-only positive-lag phase boundary: under the stated iid scaling, K and $M_s(0)$ contribute a seed-dependent leading sign.

Theorem 6 is a partial predictor: realized $d(0)$ gives an exact local sign in its stated dense-linear scope, not a global trajectory or a parameter-only law in ρ and lag. Where a higher local jet is proved—as in the zero-lag result with $d(0) = 0$ and $d'(0) < 0$ —it likewise supplies a rigorous local sign only in that stated scope. Stronger positive-lag trajectory prediction requires independently established product-ball or kernel-movement control, for which this paper supplies no nonvacuous general constants.

The proofs are in Appendix K. Equation (11) is exact for the zero-background-noise linear synthetic model. For tanh/GRU probes, we use Proposition 1's direct autograd drift as primary and report R ; projected and matched-state decompositions are diagnostics only.

C SEPARATING RATE SUPPRESSION, OUTCOME SUPPRESSION, AND CAUSAL STARVATION

Proposition 7 (Cumulative drift balance). *Let Δ be absolutely continuous on $[0, H]$, with $\Delta(0) = 0$ and $d = \Delta'$ almost everywhere. Write*

$$d_+(t) = [d(t)]_+, \quad d_-(t) = [-d(t)]_+, \quad P(t) = \int_0^t d_+(u) du, \quad N(t) = \int_0^t d_-(u) du. \quad (33)$$

Then $\Delta(t) = P(t) - N(t)$ for every $t \in [0, H]$. Thus strict outcome suppression occurs by H if and only if $N(t) > P(t)$ for some $t \in (0, H]$; this statement permits arbitrary sign changes and suppression from initialization. This is deliberately a bookkeeping proposition: its contribution is the causal separation it enforces, not proof difficulty.

Corollary 8 (Single-crossing tail-area criterion). *Suppose additionally that there is $\tau_d \in (0, H)$ such that $d > 0$ almost everywhere on $(0, \tau_d)$ and $d < 0$ almost everywhere on (τ_d, H) . Define*

$$A_+ = P(\tau_d) = \Delta(\tau_d) > 0. \quad (34)$$

Then $P(t) = A_+$ and $\Delta(t) = A_+ - N(t)$ for $t \geq \tau_d$. Hence a later equality is reached by H if and only if $N(H) \geq A_+$, and strict outcome suppression occurs by H if and only if $N(H) > A_+$. If $d < 0$ on each nontrivial post-crossing interval, the later equality is unique whenever it exists.

$A \rightarrow -$ drift crossing therefore locates a local maximum of Δ ; it does not force Δ to return to zero. Pathwise attribution requires the negative area to repay all positive area already accumulated: outcome suppression begins only when accumulated suppression overtakes accumulated help. Causal starvation further requires the independent weak-only learnability gate. A failed gate blocks that certificate but does not establish that starvation is absent.

Corollary 9 (Response-crossing bounds). *Under Corollary 8, if $d(t) \leq -\kappa < 0$ on $[t_0, H]$ with $t_0 > \tau_d$, then, when the right side lies within the horizon,*

$$\tau_{\text{response}} \leq t_0 + \frac{[A_+ - N(t_0)]_+}{\kappa} \leq t_0 + \frac{A_+}{\kappa}. \quad (35)$$

If $d(\tau_d) = 0$ and $d'(t) \leq -\lambda < 0$ on $[\tau_d, \tau_d + r]$, then $N(t) \geq \lambda(t - \tau_d)^2/2$; when $A_+ \leq \lambda r^2/2$,

$$\tau_{\text{response}} \leq \tau_d + \sqrt{2A_+/\lambda}. \quad (36)$$

Theorem 10 (Conditional parameter-tube certificate). *Let $X = (\theta_B, \theta_W)$, let V be the joint paired gradient-flow field, and define $d = L_V \Delta$ and $j = L_V d$. On the radius- R product-norm ball U_R about the shared initialization, suppose uniformly that*

$$\|V(X)\| \leq S_R, \quad -\Lambda \leq j(X) \leq -\lambda < 0, \quad 0 < \lambda \leq \Lambda, \quad (37)$$

with $S_R > 0$, and set $h_R = R/S_R$. If $d_0 > 0$ and $h_R > d_0/\lambda$, there is exactly one transfer-to-rate-suppression crossing before ball exit and

$$\frac{d_0}{\Lambda} \leq \tau_d \leq \frac{d_0}{\lambda}. \quad (38)$$

If $h_R > 2d_0/\lambda$, then $\Delta(t) < 0$ for every $2d_0/\lambda < t \leq h_R$. If also the weak-only drift obeys $\dot{M}_w^W \geq \nu > 0$ uniformly, the target is nondegenerate, and

$$\beta > M_w^W(0), \quad \frac{\beta - M_w^W(0)}{\nu} < h_R, \quad (39)$$

then the weak-only member learns by reaching β at a strictly positive time within the same certified horizon, so the outcome suppression is causally gated starvation rather than suppression alone. If $\beta \leq M_w^W(0)$, the target was met at initialization; this is degenerate attainment, not learnability, and cannot certify causal starvation.

Corollary 11 (Certified finite-horizon safe transfer). *On the same ball, if instead $|j(X)| \leq J$ and $d_0 - Jh_R > 0$, then for $0 < t \leq h_R$,*

$$d(t) \geq d_0 - Jt > 0, \quad \Delta(t) \geq d_0 t - \frac{Jt^2}{2} > 0. \quad (40)$$

Thus suppression is excluded on this certified finite horizon, not for all future optimization time. The executable checker fails closed when optional premises are missing; it does not estimate $S_R, \lambda, \Lambda, \nu$, or J from sampled trajectories. No current component constructs nonvacuous general positive-lag constants.

C.1 AN ORDERING-INVARIANT RANK-ONE CRITERION

Assume deterministic positive cues $z^B = (\rho, 1)$, $z^W = (0, 1)$, $\rho > 0$, and exact two-mode realization. Define

$$u_B = \rho M_s^B + M_w^B, \quad s_B = \sigma(-u_B), \quad (41)$$

$$u_W = M_w^W, \quad s_W = \sigma(-u_W), \quad (42)$$

$$H_B = \rho G_{ws}^B + G_{ww}^B, \quad H_W = G_{ww}^W, \quad (43)$$

$$Q_B = \rho^2 G_{ss}^B + 2\rho G_{sw}^B + G_{ww}^B \geq 0. \quad (44)$$

Theorem 12 (Log drift-ratio identity). *The weak drifts factor as $F_w^B = s_B H_B$ and $F_w^W = s_W H_W$. Where $H_B, H_W > 0$, define*

$$\Psi = \log \frac{F_w^B}{F_w^W} = \log \frac{s_B}{s_W} + \log \frac{H_B}{H_W}. \quad (45)$$

Then $\text{sign}(d) = \text{sign}(\Psi)$ and

$$\Psi' = -s_B(1 - s_B)Q_B + s_W(1 - s_W)H_W + \left(\log \frac{H_B}{H_W} \right)'. \quad (46)$$

Corollary 13 (Unique rate crossing). *Suppose on $[t_0, H]$ that $H_B, H_W > 0$, $\Psi(t_0) > 0$, $\Psi'(t) \leq -\kappa < 0$, and $H - t_0 \geq \Psi(t_0)/\kappa$. Then exactly one rate crossing occurs in $(t_0, H]$ and*

$$\tau_d \leq t_0 + \Psi(t_0)/\kappa. \quad (47)$$

A sufficient primitive condition is

$$s_B(1 - s_B)Q_B \geq a, \quad s_W(1 - s_W)H_W \leq b, \quad (\log(H_B/H_W))' \leq c, \quad a - b - c \geq \kappa > 0. \quad (48)$$

The bounds must be established independently to be predictive. Evaluating Ψ on a completed trajectory is certification, not prediction. Noisy cues generally break the rank-one factorization, and even a rate crossing still requires Corollary 8 for an outcome crossing.

C.2 HITTING-TIME STABILITY

Corollary 14 (Transverse first-hit stability). *Let continuous $x_N, x : [0, H] \rightarrow \mathbb{R}$ satisfy $\|x_N - x\|_\infty \leq \epsilon$, and define*

$$\tau = \inf\{t \in [0, H] : x(t) \geq \beta\}, \quad \tau_N = \inf\{t \in [0, H] : x_N(t) \geq \beta\}, \quad \inf \emptyset = \infty. \quad (49)$$

Assume x is absolutely continuous, $x(0) < \beta$, and $\tau \in (0, H)$. Choose $0 < r \leq \min\{\tau, H - \tau\}$ such that $x'(t) \geq \kappa > 0$ almost everywhere on $[\tau - r, \tau + r]$, and let

$$\eta = \beta - \max_{0 \leq t \leq \tau - r} x(t) > 0. \quad (50)$$

If $\epsilon < \eta$ and $\epsilon/\kappa \leq r$, then $\tau_N < \infty$ and

$$|\tau_N - \tau| \leq \epsilon/\kappa. \quad (51)$$

For an ordered grid $0 = t_0 < \dots < t_m = H$ of maximum mesh $h = \max_j(t_j - t_{j-1})$, let τ_N^{samp} be the first sampled time with $x_N(t_j) \geq \beta$, and let τ_N^{int} be the first hit of the piecewise-linear interpolant through the samples. If $\epsilon/\kappa + h \leq r$, both are finite and

$$|\tau_N^{\text{samp}} - \tau| \leq \epsilon/\kappa + h, \quad |\tau_N^{\text{int}} - \tau| \leq \epsilon/\kappa + h. \quad (52)$$

For a paired delay, if both reference hits are finite and each trajectory satisfies its own assumptions, then

$$|(\tau_{N,B} - \tau_{N,W}) - (\tau_B - \tau_W)| \leq \epsilon_B/\kappa_B + \epsilon_W/\kappa_W + h_B + h_W. \quad (53)$$

If either hit is infinite, the delay is censored; a common mesh h gives the special case with $2h$.

Descending crossings follow by sign reversal and localization after a strict excursion. Since the response gap has $\Delta(0) = 0$, its later downward outcome crossover must be localized to $[a, H]$ with $\Delta(a) > 0$ and treated using $x = -\Delta$ and target zero; the hit at time zero is not that later crossover. This corollary transfers an independently proved uniform trajectory limit. It does not prove that limit, first-entry semantics, the prehistory margin, or the derivative bound. Thus Theorem 5’s summed paired response bound licenses a crossing-time statement only after all of these separate guards hold; an error formula or empirical event match alone is not a hitting-time certificate.

D A SOLVABLE RECURRENCE-INVISIBLE ZERO-LAG WIDE LIMIT

Set lag separation zero and background input/noise zero in the dense linear model; let the realized recurrent matrix W be arbitrary and finite. Both cues then arrive at the final step. Starting from $h_0 = 0$, induction gives $h_t = 0$ for every state entering the final update and hence $h_T = Wh_{T-1} + Bx_{T-1} = Bx_{T-1}$. Thus W is loss-invisible, $\nabla_W L = 0$, and it remains fixed. Writing the input columns as b_s, b_w gives $M_a = c \cdot b_a$. Let $u_{ab} = b_a \cdot b_b$ and $w = c \cdot c$.

Theorem 15 (Every-width six-scalar closure). *At every width,*

$$\dot{M}_a = g_a w + \sum_b g_b u_{ab}, \quad \dot{u}_{ab} = g_a M_b + g_b M_a, \quad (54)$$

$$\dot{w} = 2 \sum_b g_b M_b, \quad G_{ab} = u_{ab} + \delta_{ab} w. \quad (55)$$

Theorem 16 (Finite-horizon deterministic wide limit). *For any finite realized recurrent matrix W , let all coordinates of b_s, b_w, c be mutually independent $\mathcal{N}(0, 1/N)$ and*

$$q_N = (M_s, M_w, u_{ss}, u_{sw}, u_{ww}, w), \quad q_0 = (0, 0, 1, 0, 1, 1). \quad (56)$$

For every $\epsilon > 0$,

$$\mathbb{P}(\|q_N(0) - q_0\|_\infty > \epsilon) \leq \min \left\{ 1, \frac{9}{N\epsilon^2} \right\}. \quad (57)$$

Fix a cue law with bounded second moments and define

$$g_a(M) = \mathbb{E}[z_a \sigma(-z \cdot M)], \quad A = \max_a \mathbb{E}|z_a|, \quad C = \frac{1}{4} \max_a \sum_b \mathbb{E}|z_a z_b|. \quad (58)$$

Then $\|g(M)\|_\infty \leq A$ and

$$\|g(M) - g(\widetilde{M})\|_\infty \leq C \|M - \widetilde{M}\|_\infty. \quad (59)$$

Let $q(t)$ be the deterministic solution from q_0 , let $c_\star = W(e^{-1}) = \sup_x x\sigma(-x)$, where W is Lambert's function, and set

$$R_H = 3 + 4c_\star H, \quad B_{H,r} = R_H + r, \quad L_{H,r} = 4(A + B_{H,r}C) \quad (60)$$

for a fixed deterministic radius $r > 0$. The reference Gram energy $R = w + u_{ss} + u_{ww}$ obeys $R(0) = 3$, $R' \leq 4c_\star$, and therefore $\|q(t)\|_\infty \leq R_H$ on $[0, H]$. The vector field is $L_{H,r}$ -Lipschitz in infinity norm on the radius- r tube about this path. If

$$\|q_N(0) - q_0\|_\infty \leq e^{-L_{H,r}H} \min\{r, \delta\}, \quad (61)$$

then q_N does not leave the tube before H and

$$\sup_{t \leq H} \|q_N(t) - q(t)\|_\infty \leq \delta. \quad (62)$$

Consequently,

$$\mathbb{P}\left(\sup_{t \leq H} \|q_N(t) - q(t)\|_\infty > \delta\right) \leq \min\left\{1, \frac{9e^{2L_{H,r}H}}{N \min\{r, \delta\}^2}\right\}. \quad (63)$$

For $r = 1$ and $0 < \delta \leq 1$, the denominator reduces to $N\delta^2$ with the deterministic constant $L_{H,1}$. Thus $q_N \rightarrow q$ uniformly on fixed horizons in probability, with initialization scale $O_{\mathbb{P}}(N^{-1/2})$. For the paired both/weak-only result, use a common r and the maximum of the two deterministic $L_{H,r}$ constants for the two fixed cue laws.

This theorem is a genuine large-width result at recurrence-invisible zero lag for any finite realized recurrent gain. It is not a positive-lag recurrent DMFT.

Corollary 17 (Zero-lag suppression and weak-only learnability). *For the noiseless positive rank-one cue law ($\text{cue_noise} = 0$), zero lag, zero background noise, and any finite realized recurrent matrix, use the deterministic symmetric initial state $(M_s, M_w, u_{ss}, u_{sw}, u_{ww}, w) = (0, 0, 1, 0, 1, 1)$. For every $\rho > 0$, the both-minus-weak gap and rate satisfy*

$$\bar{\Delta}(0) = 0, \quad \bar{d}(0) = 0, \quad \bar{d}'(0) = -\rho^2/2 < 0. \quad (64)$$

Hence $\bar{\Delta}(t) < 0$ for all sufficiently small $t > 0$: suppression begins at initialization and is not transfer-then-suppression. In the weak-only subsystem,

$$u_{ww} = w = \sqrt{1 + M_w^2}, \quad \dot{M}_w = 2\sigma(-M_w)\sqrt{1 + M_w^2} > 0, \quad (65)$$

so every finite $\beta > 0$ is reached and

$$\tau_\beta \leq \frac{\beta}{2\sigma(-\beta)}. \quad (66)$$

These identities and the target bound are independent of the arbitrary finite realized W . The finite-width outcome-suppression/learnability conclusion follows from Theorem 16 only after deterministic witness margins and its common tube event are established; ordinary floating-point integration is not silently promoted to such a proof.

E LOCAL MECHANISM-LEVEL CONTROL WITH COUNTERFACTUAL DRIFT CORRECTION

CDC uses the paired weak-only drift target to intervene on the diagnosed local mechanism; it is not presented as a general optimizer or robustness method. Flatten the implemented trainable tensors into one fixed parameter chart with the standard product Euclidean/Frobenius inner product. Every gradient, dot product, norm, projection, Lipschitz ball, and straight segment in this section uses that chart and metric; no reparameterization-invariance or natural-gradient claim is made.

At a both-feature state θ , let

$$a = \nabla m_s(\theta), \quad p = \nabla m_w(\theta), \quad v = -\nabla L(\theta), \quad F_s = a \cdot v, \quad F_w = p \cdot v. \quad (67)$$

A same-initialization weak-only shadow supplies F_w^W . Define

$$q = \begin{cases} p - \frac{p \cdot a}{\|a\|^2} a, & a \neq 0, \\ p, & a = 0, \end{cases} \quad \delta = [F_w^W - F_w]_+, \quad \alpha = \begin{cases} \delta / \|q\|^2, & \delta > 0, q \neq 0, \\ 0, & \delta = 0. \end{cases} \quad (68)$$

The corrected velocity is $v' = v + \alpha q$. A positive deficit with $q = 0$ is infeasible. The theorems assume full-batch gradients, plain SGD, zero weight decay, and no clipping. A nonbinding coefficient cap leaves the uncapped result unchanged; a binding cap preserves orthogonality but generally loses target attainment, so no target-attaining optimality claim then applies. In the fixed implemented chart, feasible uncapped target-attaining CDC is the unique minimum- Euclidean-norm correction satisfying zero instantaneous strong-drift change and the weak-drift lower bound. Its finite-step guarantee is local and one-step.

Theorem 18 (CDC local guarantees). *The construction in Eq. (68) has the following fixed-coordinate properties.*

1. **Instantaneous preservation:** $a \cdot v' = a \cdot v$ exactly, including when α is capped.
2. **Minimum Euclidean norm:** for the feasible uncapped problem, $d^* = \alpha q$ is the unique minimum-norm, lower-bound-attaining correction satisfying $a \cdot d = 0$ and $p \cdot (v + d) \geq F_w^W$.
3. **Local finite-step bound:** if ∇m_s is L -Lipschitz in the same norm on a convex neighborhood containing both straight coordinate step segments, then

$$|m_s(\theta + \eta v') - m_s(\theta + \eta v)| \leq \frac{L}{2} \eta^2 (\|v'\|^2 + \|v\|^2). \quad (69)$$

For the active uncapped correction, if $\|v\| \leq V$, $\delta \leq D$, and $\|q\| \geq q_{\min} > 0$, the right side is at most

$$\frac{L}{2} \eta^2 [V^2 + (V + D/q_{\min})^2]. \quad (70)$$

This explicit norm bound also remains valid under a nonnegative upper cap, which can only reduce α .

For a positive raw deficit, exact feasibility can be stated intrinsically only as dm_w having a nonzero restriction to $\ker(dm_s)$; in this metric it is equivalent to $q \neq 0$. The represented q , its norm, the numerical feasibility_epsilon, and the selected minimum-norm correction are metric- and parameter-scale-dependent. The finite-step result is one-step and local in the same chart. It is not a trajectory guarantee and does not imply final strong-response retention.

F EXECUTABLE SCOPE GUARDS

Appendix Table 7 maps theorem objects to their implementations and explicit scope guards. Helpers enforce only represented scalar or configuration conditions; callers still establish trajectory, model, first-entry, and smoothness hypotheses. Unit tests check software identities and edge cases, not mathematical propositions or unrepresented assumptions.

Every general positive-lag solver path that would require an unproved closure raises `NotImplementedError`; arbitrary finite recurrent gain is accepted only on the proved zero-lag recurrence-invisible path. This is a scientific guard, not missing numerical polish.

G EXPERIMENTAL PROTOCOL

G.1 SYNTHETIC TASK AND PAIRING

Sequences have two input channels. A strong cue of scale ρ and a unit weak cue appear at controlled sequence positions separated by a lag. Positive tasks align the cues with the label; the negative control reverses the strong cue. Data and initialization are paired across both-feature and weak-only conditions. Training is deterministic full-batch cross-entropy with zero weight decay. Except for E-NL, the two conditions can be run sequentially because deterministic full-batch updates preserve

the same trajectories; E-NL uses lockstep models so both exact drifts are available at each logged time.

Appendix Table 8 records model, task, seed, and compute settings, including the roughly doubled per-step model work required by matched- shadow methods. E1 uses four strengths and seed IDs 4–11, so each lag contains 32 ρ -seed cells but only eight independent seed IDs. Its gain is 0.5, $\beta = 0.25$, and horizon is $H = 5$; the negative control is one eight-seed cell at $(\rho, \text{lag}) = (4, 4)$. E2 uses gain 0.5 and a held-out empirical closure reference. E-NL uses one cell ($\rho = 4$, lag 2), eight seeds per architecture, gain 0.8, cue noise 0.1, zero background noise, $\beta = 0.5$, and direct autograd drift. The frozen-kernel studies use the same task cell. Historical E3 compares ERM, Spectral Decoupling, an interaction penalty, CDC, loss-gradient projection, unconstrained rescue, Bloop-style shadow-target rescue, and PCGrad-style shadow-target rescue; the five rescue arms share the same weak-only shadow target. These style labels do not assert canonical source parity.

Prospectively frozen gate and mechanism-control extensions. E26 executes 64 crossed records (four data seeds by eight model seeds per dataset) on MNIST 3-vs-8 and FashionMNIST 0-vs-6. It retains real core pixels and ablates only an exactly removable generated cue channel, so its publication scope is semi-real rather than natural-spurious-correlation evidence. E27 executes 192 nonlinear records over tanh/GRU, three fixed (ρ, lag) cells, and the prospectively frozen first-hit grid $\beta \in \{0.25, 0.5, 1, 2\}$. E28 executes 256 method-records over ERM, CDC, Bloop-style shadow-target rescue, and PCGrad-style shadow-target rescue. The latter two are response-gradient shadow-target variants with `canonical:false`; no canonical algorithm/reference parity is claimed. All three studies were externally sealed and executed from clean commit 9ebbc617a9aaad5c3c19e9d5fd4fa08f2d3517d0. The excluded evaluator- engineering set and its non-scientific use are documented in Appendix M.

Manifest-sealed surrogate audit protocol. `enl-preflight` constructs no optimizer and observes no trained state. It writes the initialization kernels and predictions, verifies state/gradient nonmutation, and hashes every declared file and per-record kernel together with the resolved config and executable-source fingerprint. Before constructing an optimizer, `enl-evaluate` requires the sidecar, actual manifest, and externally pinned digest to agree; verifies every hash/size, source, protocol, seed pair, and time grid; and loads rather than rebuilds predictions. It re-hashes all sealed inputs after training. The broad five-endpoint rule was frozen before its pilot; after that rule failed, only drift/response crossing was frozen as primary on fresh seeds. The latter crossed design uses a two-way pigeonhole bootstrap with 10,000 replicates.

Statistics and censoring. Independent seed pairs are the unit of replication only when neither seed axis is reused. E1’s 32 cells per lag reuse eight seed IDs across four strengths, and the E-NL 3/8, 5/8, and 0/8 proportions describe only the eight observed seeds per architecture; none is an estimate of universal event prevalence. Under E20’s crossed data/model-seed reuse, row-wise Student- t intervals are suppressed and the two-way percentile interval resamples the observed seed axes independently. It is conditional on those observed levels: an all-correct table necessarily gives $[1, 1]$ and does not quantify unseen failure modes. Hitting times beyond the horizon are right-censored and never converted into finite delays. Conditional crossing-time means are reported only with their crossing-record denominator. The exploratory E-NL 5×5 grid uses seeds 0–3 and is explicitly post-hoc; it is not used to select a confirmatory condition. E26 reports dataset-specific two-way crossed-seed intervals; absent weak-only first hits censor causal interpretation in both directions. E27 uses a shared 5,000-replicate two-way pigeonhole bootstrap, equal-weights the three fixed cells, and Holm-adjusts the two architecture tests; architecture ranking was forbidden. E28 uses the same bootstrap draws and equal-weights its two fixed cells. Its weak-rescue contrast is CDC minus comparator for $-|\text{weak AUC gap}|$ (positive favors CDC); its trajectory and final contrasts are comparator minus CDC absolute deviation from paired ERM (positive favors CDC).

Reproducibility. Ordinary runs write resolved configuration, environment, trajectory, summary, and figure artifacts; frozen studies additionally seal predictions and hashes before training. The final E-NL run archives a 17-file executable-source fingerprint and unmodified metadata, making that named run reconstructible. E19 and E20 are different: their manifests, predictions, scores, seals, and historical source fingerprints survive, but the exact dirty Python source bytes were not archived and are unavailable and unreconstructible. Current source intentionally fails their source guard; a

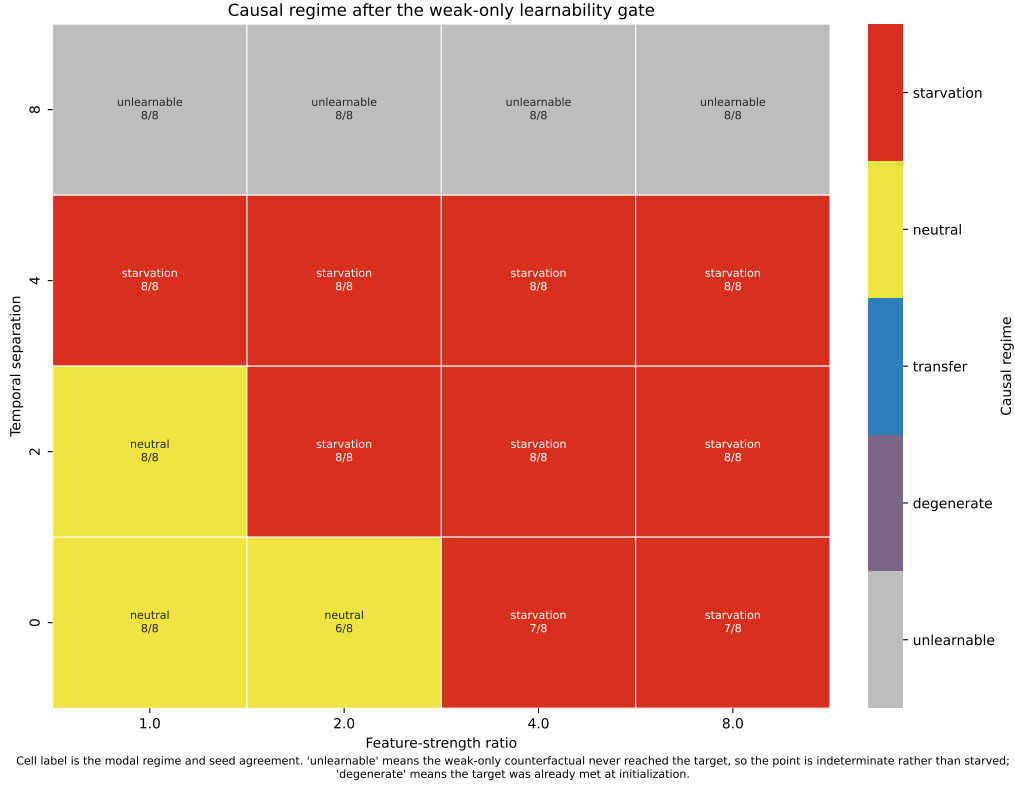


Figure 2: Controlled dense-linear paired weak-AUC deficits and weak-only gate status. The specified learnability gate makes all lag-8 points causally indeterminate rather than positive or negative starvation evidence.

new preflight/evaluation is a new study, not a replay. Historical E1, E2, E2-R, exploratory E-NL, and E3 artifacts likewise disclose dirty states predating source fingerprinting. By contrast, E26–E28 were executed from clean commit 9ebbc617a9aad5c3c19e9d5fd4fa08f2d3517d0; their source manifests make the source reconstructible. The compact E26 archive omits 192 records/** files and the shared E27/E28 archive omits 1,792, so hashes and central outputs remain auditable but neither compact archive is a standalone replay package.

H RESULTS

H.1 PAIRED WEAK-AUC DEFICITS AND WEAK-ONLY LEARNABILITY

At lags 0, 2, and 4, all 32 ρ -seed cells per lag (four strengths by eight seed IDs) have positive paired weak-AUC deficits and learnable weak-only controls under the specified gate. Averaged across ρ , the lag-wise deficits are 5.76, 8.93, and 6.11, respectively; the measured effect peaks at lag 2 rather than decreasing monotonically with lag. At every fixed lag, the mean deficit increases monotonically over the measured ρ grid. The negative-control cell is transfer in 8/8 seeds, with mean gap -4.038 and 95% CI $[-4.498, -3.578]$. Every lag-8 cell fails the weak-only gate: all 32 are weak-only nonlearnable and causally indeterminate under $\beta = 0.25$, $H = 5$. They supply neither causal-starvation certificates nor evidence that starvation is absent. Final accuracy is 1.0 throughout, and no row meets the target at initialization.

H.2 EMPIRICAL WIDTH TRENDS AND EXACT ZERO-LAG CHECKS

E2 compares finite networks with a disjoint-seed empirical *closure reference*; it is not an independently derived theory. Overall both-condition trajectory NRMSE is 0.09084, 0.04761, 0.05149, 0.02958 at

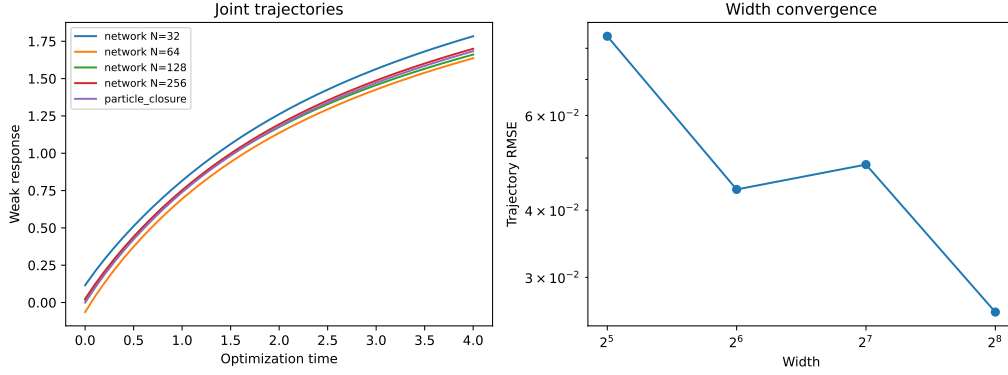


Figure 3: Finite-width agreement with a held-out empirical finite-network closure reference. This is not DMFT validation.

Table 3: Final E-NL result at $\rho = 4$, lag 2, eight seeds. Crossing times and confidence intervals are conditional on crossing and must not be read as eight-seed estimates.

Model	exact drift	response	causal	mean τ_d	mean τ_r	max $ R_w^B - R_w^W $
tanh	3/8	3/8	3/8	0.194	0.389	0.02197
GRU	5/8	5/8	0/8	3.846	7.200	0.002986

widths 32, 64, 128, 256. Weak-only NRMSE is 1.56931, 0.38792, 0.40201, 0.20024. Both improve end-to-end but are non-monotone at $N = 128$. All 32 point-condition-diagnostic combinations improve end-to-end; large normalized weak-only values at difficult points have near-zero denominators. The exact finite-width geometry check has zero recorded relative error and the maximum projected-identity error is 2.86×10^{-6} .

In the exact legacy-named zero-disorder solver check, the time-discretization residual scales as $O(\eta)$ with fitted slope 1.0115. A frozen-geometry integrator agrees with the projected-flow integrator to 1.91×10^{-15} . The general solver acceptance record remains false by construction because all general-DMFT checks are blocked.

H.3 TANH CERTIFICATES VERSUS UNGATED GRU CROSSINGS

Table 3 is the authoritative nonlinear result at the fixed $(\rho, \text{lag}) = (4, 2)$ cell. Tanh has a complete exact-drift/response/tail- area transfer-to-starvation certificate in 3/8 observed seeds; the other five have negative exact relative drift from initialization and therefore do not exhibit the transfer-to-starvation path type. GRU has exact drift and response crossings in 5/8 seeds, but every weak-only trajectory fails to reach $\beta = 0.5$; its causal count is therefore 0/8 and causal status is indeterminate under that gate. This separates rate/outcome suppression from causal starvation; it does not rank architectures or estimate prevalence. The nonzero direct-versus-projected residuals empirically confirm that Gg cannot be silently treated as the exact nonlinear probe derivative.

The preregistered gate is the first hit at $\beta = 0.5$, but a single binary operating point can hide whether failures are marginal. We therefore add a post- hoc terminal audit using $B_{W,s}(H) = M_{w,s}^W(H)$ and the empirical fraction of seeds with $B_{W,s}(H) \geq \beta$ over candidate thresholds. No model was retrained: Figure 1 is derived from the tracked per-seed E-NL summary. Tanh terminal min/mean/max are 4.41313266754/4.49664855003/4.67605876923, with 8/8 above $\beta = 0.5$; GRU min/mean/max are 0.0828229486942/0.129690139554/0.187142759562, with 0/8 above 0.5. Thus the GRU weak-only gate failure is not marginal at that operating point, but it still provides neither a causal-starvation certificate nor evidence that starvation is absent. Terminal reach is not generally interchangeable with the preregistered first-hit event on a nonmonotone trajectory, and this post-hoc diagnostic supports neither a new confirmatory endpoint nor an architecture ranking.

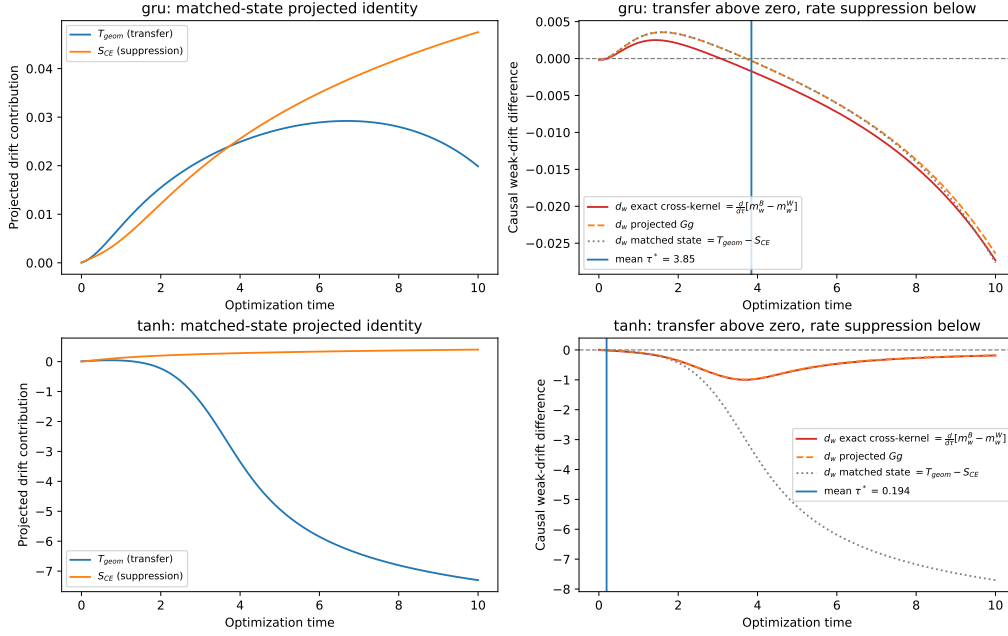


Figure 4: Paired nonlinear diagnostic trajectories. Direct-autograd drift is primary; projected and matched-state quantities are separately styled diagnostics.

Table 4: Manifest-sealed frozen-kernel evaluations. The broad pilot’s complete five-output rule failed. Only the two event rows in the fresh factorial were preregistered primary endpoints. CIs resample data- and model-seed axes.

Protocol	Endpoint	Overall	tanh	GRU	95% CI
Broad pilot	phase	12/16	6/8	6/8	diagnostic
Broad pilot	weak learnability	12/16	4/8	8/8	failed
Restricted 4×8	drift event	60/64	32/32	28/32	$[0.8125, 1]$
Restricted 4×8	response event	64/64	32/32	32/32	$[1, 1]$

The post-hoc tanh search over $\rho \in \{1.5, 2, 3, 4, 6\}$ and lags $\{0, 1, 2, 4, 8\}$ found no cell with 4/4 causal certificates; the best cells reached 3/4. We retain this as a negative search audit and do not select a searched cell for a paper claim. In particular, the superseded 8/8 tanh result and its $\tau^* = 1.6112$ used a different response convention and are not valid under the final definition.

H.4 E19 FALSIFICATION; E20 RESTRICTED EVENT-LABEL FOLLOW-UP

The initialization-frozen tangent surrogate was first tested under a sealed five-output pilot using one unseen data seed and eight unseen model seeds per architecture. It obtained 12/16 phase, 14/16 drift-crossing, 14/16 response-crossing, 14/16 causal- certificate, and 12/16 weak-learnability classifications. The preregistered rule nevertheless *failed*: tanh learnability was 4/8, below the 5/8 per-model floor. We preserve that rejection rather than averaging it away.

Only after this failure did we define the narrower crossing-event hypothesis. A fresh sealed factorial crossed four unseen data seeds with eight unseen model seeds for each architecture. Table 4 reports classification correctness, not observed crossing prevalence. The restricted rule passed: all response-crossing labels and 60/64 drift-crossing labels were correct; the two-way bootstrap lower endpoints exceeded the frozen 0.5 floor.

The restriction is substantive. In the factorial, phase, causal-certificate, and learnability correctness were only 52/64, 56/64, and 49/64 and were explicitly secondary; tanh learnability was 17/32. Crossing times remained systematically early (overall drift/response bias $-0.822/-1.189$), and

Table 5: Dense CDC ablation, eight paired seeds. Shadow-based arms have access to a matched weak-only model; the other arms do not. “Strong change” is the maximum absolute instantaneous strong-drift perturbation.

Method	Shadow	Weak AUC gap	Strong change	Final m_s
ERM	no	10.5155	—	1.364
Spectral Decoupling	no	9.0229	—	0.829
Interaction	no	10.5145	—	1.365
CDC	yes	−0.0005	4.77×10^{-7}	1.042
Loss-gradient projection	yes	−0.0002	3.75×10^{-1}	0.371
Unconstrained rescue	yes	−0.0007	1.48×10^{-1}	1.295
Bloop-style shadow-target rescue	yes	0.0002	3.75×10^{-1}	0.370
PCGrad-style shadow-target rescue	yes	−0.0007	1.48×10^{-1}	1.295

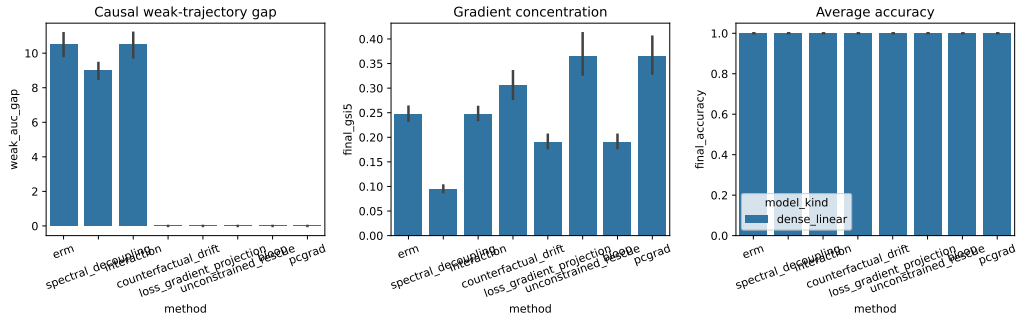


Figure 5: Eight-method dense-linear mitigation ablation. CDC’s established distinction is instantaneous first-order preservation, not superior weak rescue or final strong retention.

tanh response-gap and weak- response RMSE were 2.455 and 2.489 despite high sign agreement. Thus the evidence supports an uncalibrated event classifier at this one width/task cell. It neither proves Theorem 5’s kernel premise nor validates phase, causal starvation, learnability, trajectory magnitude, calibrated time, or prevalence. Appendix O reports the complete failure, restricted-success, timing-bias, and magnitude-error numbers.

H.5 HISTORICAL E3 CDC ABLATION: LOCAL MECHANISM CONTROL, NOT BEST FINAL TRADEOFF

In historical E3, all five shadow-based methods reduce the causal weak AUC gap from ERM’s 10.5155 to approximately zero and attain their shared weak-drift target at every logged step. Their weak outcomes are numerically similar to roughly four decimal places, but not statistically equivalent: 9/10 direct paired comparisons are significant at 0.05, while the differences are practically tiny relative to the roughly 10.5 change from ERM. CDC alone preserves instantaneous strong drift to rounding scale, 4.77×10^{-7} , versus 1.48×10^{-1} to 3.75×10^{-1} for the alternative shadow constraints. This local property does not yield the best final strong response: unconstrained rescue and the PCGrad-style shadow-target rescue end at 1.295, compared with CDC’s 1.042 and ERM’s 1.364. These are historical E3 outcomes, distinct from E28’s later frozen tradeoff.

H.6 CAUSAL GATE INDETERMINACY AND COMPARATOR-RESTRICTED MECHANISM CONTROL

E26 completed 64 semi-real records. MNIST 3-vs-8 and FashionMNIST 0-vs-6 each had 0/32 weak-only-learnable records and therefore 0/32 causal certificates under the specified gate. Their mean weak-AUC gaps were as follows. For MNIST 3-vs-8, the gap was $-1.2362842136667493 \times 10^{-3}$. Its 95% CI was $[-3.4405428466004646, 1.0159591371180453] \times 10^{-3}$. For FashionMNIST 0-vs-6, the gap was $-1.0921728159018996 \times 10^{-4}$. Its 95% CI was $[-4.129384520886106, 3.182670049955049] \times 10^{-3}$. Positive means weak-only core gain exceeds both-feature core gain.

Table 6: E28 equal-weight two-cell macro contrasts over 256 method-records. Both comparators are non-canonical. Brackets are 95% bootstrap CIs.

Comparator	CDC-comparator weak rescue	Comparator-CDC trajectory dev.	Comparator-CDC final dev.	Joint
Bloop-style shadow-target rescue	0.00219825939210656 [0.0009770187294930126, 0.003277306207946477]	3.4651361284978416 [3.3443827215131754, 3.5849713202780952]	0.6422362388111651 [0.6286088704226539, 0.6560108084365315]	true
PCGrad-style shadow-target rescue	-0.00015351238407574523 [-0.00047905348592839755, 0.000352480124139808214]	-0.46708880128076556 [-0.4887378844428417, -0.27378367737439074]	-0.2565436447720421 [-0.3016841153614223, -0.208765183574514]	false

The frozen hypothesis failed. Because every weak-only first-hit gate failed, the causal gate correctly rejects a false-positive starvation interpretation, but the zero causal counts remain negative/indeterminate: they are not evidence that starvation is absent and do not show that either dataset is intrinsically unlearnable.

E27 completed 192 nonlinear records. The equal-weight three-cell all- β causal-certificate macro estimate was 0.2916666666666667 for tanh (95% CI [0.16666666666666666, 0.4166666666666667]) and 0 for GRU (95% CI [0, 0]); the Holm-adjusted one-sided p -value was 1.0 for each. The overall beta-robust claim is false. Architecture ranking was forbidden and neither performed nor supported.

E28 completed 256 method-records. Define weak rescue as $-|\text{weak AUC gap}|$, so the CDC-minus-comparator contrast is higher-better. Define trajectory and final deviation as absolute deviations from paired ERM, so comparator-minus-CDC is positive when CDC preserves the strong response better. Table 6 reports every frozen macro contrast. The frozen joint weak-rescue/strong-preservation tradeoff is true only versus Bloop-style shadow-target rescue. It is false versus PCGrad-style shadow-target rescue, so CDC does not beat that variant under the frozen joint criterion. Both comparators have `canonical: false`; these results establish neither SOTA performance, canonical parity, nor broad CDC superiority.

E20 remains the submission-primary empirical study, with only drift- and response-crossing classification primary. E26–E28 were prospectively frozen but completed later; their outcomes do not reprioritize E20 or promote its secondary phase, causal, learnability, timing, or magnitude outputs.

I DISCUSSION AND LIMITATIONS

Causal diagnosis and empirical findings. Weak learning failure alone is not gradient starvation. E1 shows positive paired weak-AUC deficits with learnable weak-only controls in all 32 ρ -seed cells per lag at lags 0, 2, and 4, but those cells reuse eight seed IDs and do not imply a universal law. All 32 lag-8 cells instead fail the gate under $\beta = 0.25$, $H = 5$, making their causal status indeterminate. At E-NL’s fixed $(\rho, \text{lag}) = (4, 2)$ cell, tanh has complete transfer-to-starvation certificates in 3/8 observed seeds; GRU has drift and response crossings in 5/8 but 0/8 causal certificates because every weak-only path fails $\beta = 0.5$. The post-hoc terminal profile shows that this GRU gate failure is not marginal at that operating point, but it neither replaces the first-hit gate nor ranks architectures. These counts are descriptive, not prevalence estimates, and the exploratory search found no 4/4 robust cell. E26 adds semi-real breadth but is negative/indeterminate because MNIST 3-vs-8 and FashionMNIST 0-vs-6 each have 0/32 weak-only-learnable and 0/32 causal records; the gate blocks a false-positive starvation label but cannot establish absence or intrinsic dataset unlearnability. E27 rejects the frozen all- β claim for both architectures without licensing a ranking. E28 is mixed and comparator-restricted: its frozen joint tradeoff holds against the non-canonical Bloop-style shadow-target rescue but not the non-canonical PCGrad-style shadow-target rescue. None of these later outcomes changes E20’s formal submission-primary drift/response-only scope.

Exact dynamics and pathwise attribution. The exact finite-width identity $\dot{M}_w = c_w^\top q(r)/n$ separates cross-entropy sample weighting $q(r)$ from response-specific geometry c_w with no DMFT assumption. Together with exact sample-logit dynamics, dense-linear geometry, and the matched intervention, it identifies where a weak-response rate difference comes from without silently imposing a two-response closure. The cumulative balance then gives pathwise attribution for arbitrary sign changes: $\Delta = P - N$, so outcome suppression occurs only when accumulated suppression overtakes accumulated help. Neither rate nor outcome suppression is by itself a causal-starvation certificate; independently demonstrated weak-only learnability is still required.

Partial prediction and formal scope. The realized-initialization $d(0)$ criterion gives an exact local sign in its stated dense-linear setting. Where a higher local jet is proved, such as $d(0) = 0$, $d'(0) < 0$

in the recurrence-invisible zero-lag result, it likewise gives a rigorous local sign in that scope. The conditional parameter-tube and rank-one criteria, transverse hitting-time bound, recurrence-invisible zero-lag wide limit, and fixed-chart CDC result are mathematical statements with proofs, but their scopes differ. The cumulative balance characterizes outcome suppression without supplying weak-only learnability; the rank-one criterion addresses rate crossing; the wide limit requires zero lag and zero background input, although it permits any finite realized recurrent matrix. Stronger positive-lag parameter-level prediction from ρ , lag, and recurrent geometry remains unresolved because no nonvacuous general product-ball or kernel-movement constants are constructed.

Frozen-surrogate sequence. The initialization-frozen tangent surrogate retains nonlinear cross-entropy but is exact only for its tangent model. Its comparison with trained tanh/GRU networks assumes unproved instantaneous or secant-kernel movement control. E19 failed its preregistered five-endpoint rule; correctness on other labels cannot rescue the tanh learnability failure. E20 was designed only after that rejection and supports a narrower claim: drift/response crossing-event classification on one fresh crossed-seed factorial. Its phase, causal, and learnability labels were secondary, its time predictions were systematically early, and tanh trajectory magnitudes were poor. Manifest integrity proves that frozen inputs were consumed unchanged; it does not prove kernel stability, surrogate accuracy, or general causal prediction.

Local mechanism-level control and baselines. In the fixed implemented chart, feasible uncapped target-attaining CDC is the unique minimum-Euclidean-norm correction satisfying zero instantaneous strong- drift change and the weak-drift lower bound; its finite-step guarantee is local and one-step. Historical E3 exhibits that instantaneous distinction within its tested shadow-method family, but does not show a general practical advantage: weak rescue is numerically similar across shadow arms, other arms retain a larger final strong response, and all shadow methods use stronger matched-counterfactual information. E28 passes the separately frozen joint weak-rescue/strong-preservation criterion only against Bloop-style shadow-target rescue; CDC does not beat PCGrad-style shadow-target rescue under that criterion. Both comparators are non-canonical response-gradient shadow-target variants, not independently audited canonical replicas. CDC assumes a constructible weak-only shadow and roughly doubled model work; no SOTA, canonical parity, or broad superiority follows.

External validity. The evidence covers controlled synthetic task families and one semi-real generated- cue intervention over fixed cells and seed grids. E26 is not an unmodified natural shortcut or natural-spurious-correlation study. E27/E28 broaden the fixed synthetic cells and methods but do not establish architecture rankings, prevalence, canonical baseline parity, or broad CDC superiority. The unrun Waterbirds adapter is a minibatch head-coordinate surrogate outside the full-batch matched-shadow theorem, and its intercept is not an identified bird-shape response. We make no Waterbirds or natural-spurious-correlation claim. General recurrent causal prediction and calibrated tanh/GRU crossing times remain open.

Provenance. The final E-NL source is reconstructible through its archived 17-file executable- source fingerprint and preserved metadata. E19 and E20 preserve compact outcomes, seals, and source digests, but not the exact dirty source bytes; they are auditable but not historically replayable. Earlier E1, E2, E2-R, exploratory E-NL, and historical E3 share a disclosed dirty-source limitation. E26–E28 instead ran from clean commit 9ebbc617a9aaad5c3c19e9d5fd4fa08f2d3517d0; their recorded source manifests make source reconstruction possible. The E26 compact archive omits 192 records/** files and the combined E27/E28 archive omits 1,792. Their committed hashes and central files support audit, but omitted bytes cannot be recovered from hashes, so the compact archives are not standalone replay packages.

J CONCLUSION

Weak learning failure alone is not gradient starvation. A small weak response can reflect failure of the weak-only control to meet the specified target, altered cross-entropy weighting, response-specific geometry, rate suppression that has not yet reversed earlier help, or causally gated outcome suppression. The matched same-initialization intervention separates these possibilities. Exact finite- width dynamics give $\dot{M}_w = c_w^\top q(r)/n$ without a DMFT assumption; the pathwise identity

$\Delta = P - N$ then shows that outcome suppression begins only when accumulated suppression overtakes accumulated help. Causal starvation additionally requires an independently learnable weak-only counterfactual. A failed gate licenses neither a starvation certificate nor evidence that starvation is absent; causal status is indeterminate under the specified target and horizon.

Prediction remains partial. Realized $d(0)$ and proved higher local jets give rigorous local or assumption-scoped sufficient signs. The conditional rank-one, parameter-tube, hitting-time, and recurrence-invisible zero-lag results state their additional assumptions explicitly; the noiseless symmetric zero-lag reference starts suppressed, permits arbitrary finite recurrent gain, and does not supply a transfer phase. Because nonvacuous general positive-lag product-ball and kernel- movement constants are unavailable, the paper does not provide a global predictor from ρ , lag, and RNN geometry.

The empirical sequence supports the diagnostic separation rather than a universal starvation law. Corrected E1 has positive paired weak-AUC deficits and learnable weak-only controls across all 32 ρ -seed cells per lag at lags 0, 2, and 4, not 32 independent seeds; every lag-8 cell fails the gate under $\beta = 0.25$, $H = 5$ and is causally indeterminate. At E-NL's fixed $(\rho, \text{lag}) = (4, 2)$ cell, tanh has complete transfer-to-starvation certificates in 3/8 observed seeds, whereas GRU has drift and response crossings in 5/8 but 0/8 causal certificates because every weak-only path fails $\beta = 0.5$. The post-hoc terminal profile strengthens that operating-point diagnosis without replacing the first-hit gate, ranking architectures, or estimating prevalence. E19 remains a failed broad frozen-kernel pilot. E20 remains the formal submission-primary empirical study and supports only restricted frozen drift/response crossing-event labels. E26 is a completed negative/indeterminate semi-real study: MNIST 3-vs-8 and FashionMNIST 0-vs-6 each have 0/32 weak-only learnability and 0/32 causal records, so the gate blocks a false-positive label but cannot show absent starvation or intrinsic dataset unlearnability. E27's all- β claim is negative for both architectures without a ranking.

As mechanism-level control, in the fixed implemented chart, feasible uncapped target-attaining CDC is the unique minimum-Euclidean-norm correction satisfying zero instantaneous strong-drift change and the weak-drift lower bound; its finite- step guarantee is local and one-step. Historical E3 demonstrates that local constraint but not broad practical superiority or best final retention. E28's frozen joint weak-rescue/strong-preservation tradeoff succeeds only against the non-canonical Bloop-style shadow-target rescue; CDC does not beat the non-canonical PCGrad-style shadow-target rescue under that criterion. Thus no SOTA, architecture ranking, canonical parity, or broad CDC superiority follows.

The strongest supported chain is exact dynamics $\rightarrow$ matched causal intervention $\rightarrow$ pathwise attribution $\rightarrow$ local mechanism-level control. With fixed seed grids, no natural-spurious-correlation benchmark, historically unreconstructible source for E19/E20, compact E26–E28 archives that are not standalone replay packages, and no general positive-lag DMFT, general recurrent causal prediction and calibrated crossover times remain open.

K PROOFS FOR EXACT DYNAMICS AND CAUSAL-SEPARATION RESULTS

K.1 PROOF OF PROPOSITION 1 AND COROLLARY 2

For $\ell(r) = \log(1 + e^{-r})$, $\ell'(r) = -\sigma(-r)$. Therefore

$$\dot{\theta}_C = -\nabla L_C = \frac{1}{n_C} \sum_i \sigma(-r_i^C) \nabla r_i^C. \quad (71)$$

The chain rule gives

$$\dot{M}_a^C = \langle \nabla M_a^C, \dot{\theta}_C \rangle = \frac{1}{n_C} \sum_i \langle \nabla M_a^C, \nabla r_i^C \rangle \sigma(-r_i^C), \quad (72)$$

which proves Proposition 1. Under exact realization, $\nabla r_i^C = \sum_b z_{ib}^C \nabla M_b^C$. Substitution and exchange of finite sums yield Eq. (11). If an orthogonal residual logit term is present, substitution of $\nabla r_i^C = \sum_b z_{ib}^C \nabla M_b^C + \nabla r_i^\perp$ instead yields Eq. (12). $\square$

K.2 PROOF OF COROLLARY 3 AND THEOREM 5

The parameter flow above can be written $\dot{\theta} = J^\top q(r)/n$. Therefore the chain rule gives $\dot{r} = J\dot{\theta} = JJ^\top q(r)/n$ and $\dot{M} = \langle \nabla M, \dot{\theta} \rangle = c^\top q(r)/n$, proving Corollary 3.

For the comparison, q is $1/4$ -Lipschitz componentwise and hence in Euclidean norm. Subtracting Eq. (15) from Eq. (14), adding and subtracting $\Theta_0 q(r)$, and writing $e_r = \|r - \bar{r}\|$ gives

$$e'_r(u) \leq \frac{\|\Theta_0\|}{4n} e_r(u) + \frac{B_\Theta V}{n} = a e_r(u) + \frac{B_\Theta V}{n}. \quad (73)$$

Grönwall with $e_r(0) = 0$ proves Eq. (17). The same add-subtract step gives

$$|\dot{M} - \dot{M}| \leq \frac{B_c V}{n} + \frac{\|c_0\|}{4n} e_r. \quad (74)$$

Integrating and using $\int_0^t \phi_a(u) du = \psi_a(t)$ proves Eq. (18).

For SGD, the fundamental theorem of calculus along the straight update segment gives

$$r_{j+1} - r_j = \tilde{J}_j \delta \theta_j = \frac{\eta}{n} \tilde{\Theta}_j q(r_j), \quad (75)$$

$$M_{j+1} - M_j = \langle \tilde{m}_j, \delta \theta_j \rangle = \frac{\eta}{n} \tilde{c}_j^\top q(r_j). \quad (76)$$

Subtracting the frozen Euler recursion yields scalar recurrences

$$e_{j+1} \leq (1 + \eta a) e_j + \frac{\eta D_\Theta V}{n}, \quad (77)$$

$$E_{j+1} \leq E_j + \frac{\eta D_c V}{n} + \frac{\eta \|c_0\|}{4n} e_j. \quad (78)$$

Solving the first geometric recurrence and summing it in the second gives Eq. (22), including the stated $a = 0$ limits. Paired bounds are the triangle inequality. Error-separated sign transfer follows immediately; first-hit transfer requires the additional assumptions of Corollary 14. $\square$

K.3 PROOF OF THE DENSE-LINEAR GEOMETRY

For a unit response mode in channel a at lag ℓ_a , write b_a for the corresponding input column. Direct differentiation of $c^\top W^{\ell_a} b_a$ gives gradient $p_a = W^{\ell_a} b_a$ with respect to c , gradient $q_a = (W^{\ell_a})^\top c$ with respect to column b_a , and recurrent gradient

$$S_a = \sum_{j=0}^{\ell_a-1} (W^j)^\top c (W^{\ell_a-1-j} b_a)^\top. \quad (79)$$

The parameter blocks are disjoint, so their gradient inner products add. Input columns contribute only when the channels coincide, giving Eq. (24). $\square$

K.4 PROOF OF THEOREM 6

For the positive rank-one cue laws, $g^B = s_B(\rho, 1)$ and $g^W = s_W(0, 1)$ at the shared initialization. Corollary 2 and the exact dense geometry give

$$\dot{M}_w^B(0) = s_B(\rho G_{ws} + G_{ww}), \quad \dot{M}_w^W(0) = s_W G_{ww}. \quad (80)$$

Subtraction, with $K = G_{ws}$ and $A = G_{ww}$, proves Eq. (30). Since $\rho s_B > 0$, rearrangement gives Eq. (31). The displayed K, A, S_w formulas are Eq. (24) specialized to a zero-lag strong mode and an ℓ -lag weak mode. Under the iid width scaling at fixed positive ℓ , expanding the gates at zero gives

$$d_0 = \rho \left(\frac{K}{2} - \frac{A M_s(0)}{4} \right) + O_{\mathbb{P}}(N^{-1}), \quad (81)$$

whose leading sign is centered and realization-dependent. Shared initialization also gives $\Delta(0) = 0$ and $\Delta'(0) = d_0$. Therefore $\Delta(t)/t \rightarrow d_0$ as $t \downarrow 0$: a strict negative (positive) d_0 forces strict local outcome suppression (transfer) on some sufficiently small positive interval. This limit establishes no quantitative endpoint, and when $d_0 = 0$ first order is silent. This proves the exact criterion, its strongest unconditional local outcome implication, and the stated obstruction to a distribution-only positive-lag boundary. $\square$

K.5 PROOF OF PROPOSITION 7 AND COROLLARIES 8–9

The positive and negative parts satisfy $d = d_+ - d_-$ almost everywhere. Absolute continuity and the fundamental theorem of calculus therefore give

$$\Delta(t) = \Delta(0) + \int_0^t d(u) du = P(t) - N(t), \quad (82)$$

proving Proposition 7 and its pointwise sign equivalences. Under Corollary 8’s sign pattern, $d_- = 0$ before τ_d and $d_+ = 0$ afterward. Hence $P(t) = P(\tau_d) = A_+$ and

$$\Delta(t) = A_+ - N(t), \quad t \geq \tau_d. \quad (83)$$

The post-crossing sign assumption makes N continuous and strictly increasing on every nontrivial post-crossing interval. Later equality, strict negativity, and uniqueness are therefore respectively equivalent to $N(H) \geq A_+$, $N(H) > A_+$, and strict post-crossing growth of N .

If $d \leq -\kappa$ after t_0 , then $N(t) - N(t_0) \geq \kappa(t - t_0)$; solve for the time needed to cover the remaining area. If $d'(t) \leq -\lambda$ and $d(\tau_d) = 0$, integration gives $d(t) \leq -\lambda(t - \tau_d)$ and then $N(t) - N(\tau_d) \geq \lambda(t - \tau_d)^2/2$. Since $N(\tau_d) = 0$ under the single-crossing hypotheses, applying Corollary 8 proves both stated bounds. $\square$

K.6 PROOF OF THEOREM 10 AND COROLLARY 11

Before exit from U_R , path length is at most $S_R t$, so exit cannot occur before $h_R = R/S_R$. Along this interval, $d' = j$, and integration of Eq. (37) gives

$$d_0 - \Lambda t \leq d(t) \leq d_0 - \lambda t. \quad (84)$$

The lower bound is positive before d_0/Λ , the upper bound is negative after d_0/λ , continuity gives a zero between them, and $d' < 0$ makes it unique. This proves Eq. (38). Integrating the upper bound from $\Delta(0) = 0$ gives

$$\Delta(t) \leq d_0 t - \frac{\lambda t^2}{2} < 0 \quad \text{for } t > 2d_0/\lambda. \quad (85)$$

For the required nondegenerate case $\beta > M_w^W(0)$, integrating the weak-only lower drift bound reaches β by the positive time in Eq. (39), proving the causal conclusion. If $\beta \leq M_w^W(0)$, attainment occurred before optimization and supplies no learnability gate.

For the safe case, $|j| \leq J$ gives $d(t) \geq d_0 - Jt$; one more integration gives Eq. (40). The strict premise $d_0 - Jh_R > 0$ makes both quantities positive on the certified horizon. $\square$

K.7 PROOF OF THEOREM 12 AND COROLLARY 13

The deterministic cue law gives $g^B = s_B(\rho, 1)$ and $g^W = s_W(0, 1)$. Equation (11) therefore gives $F_w^B = s_B H_B$, $F_w^W = s_W H_W$, $u_B' = s_B Q_B$, and $u_W' = s_W H_W$. Since

$$\frac{d}{du} \log \sigma(-u) = -(1 - \sigma(-u)), \quad (86)$$

differentiating Ψ yields Eq. (46). When both drifts are positive, monotonicity of the logarithm gives $\text{sign}(F_w^B - F_w^W) = \text{sign}(\Psi)$.

Under $\Psi' \leq -\kappa$, integration gives $\Psi(t) \leq \Psi(t_0) - \kappa(t - t_0)$. Continuity forces a zero by $t_0 + \Psi(t_0)/\kappa \leq H$, while strict decrease gives uniqueness. The primitive inequalities imply the derivative bound term by term. $\square$

K.8 PROOF OF COROLLARY 14

Continuity and first-entry semantics imply $x(\tau) = \beta$. Absolute continuity and $x' \geq \kappa$ almost everywhere give, by integration,

$$x(t) \leq \beta - \kappa(\tau - t) \quad (\tau - r \leq t \leq \tau), \quad x(t) \geq \beta + \kappa(t - \tau) \quad (\tau \leq t \leq \tau + r). \quad (87)$$

On $[0, \tau - r]$, the quantitative prehistory margin and $\epsilon < \eta$ give $x_N < \beta$. On $[\tau - r, \tau - \epsilon/\kappa]$, the left cone and the uniform error again give $x_N < \beta$. Hence $\tau_N \geq \tau - \epsilon/\kappa$. At $t = \tau + \epsilon/\kappa$, the right cone gives $x_N(t) \geq \beta$, so the hit is finite and $\tau_N \leq \tau + \epsilon/\kappa$. This proves Eq. (51).

For sampled data, the first grid point at or after $\tau + \epsilon/\kappa$ is at most h later and, by the radius condition, remains inside the right cone; it is therefore at or above β . No sampled hit can precede the continuous lower bound. If the piecewise-linear interpolant first crosses on $[t_{j-1}, t_j]$, then $x_N(t_j) \geq \beta$, so $t_j \geq \tau - \epsilon/\kappa$ and $t_{j-1} \geq \tau - \epsilon/\kappa - h$; its upper bound is no later than the sampled hit. This proves Eq. (52). Applying the bound separately to two finite hits and using the triangle inequality proves the paired statement. $\square$

K.9 PROOF OF THEOREMS 15 AND 16

At zero lag and zero background input, every cue occurs at the final step. From $h_0 = 0$, induction gives zero hidden state before that step, so for every finite realized W ,

$$h_T = Wh_{T-1} + Bx_{T-1} = Bx_{T-1}. \quad (88)$$

The loss is independent of W , each recurrent-gradient contribution contains a zero incoming state, and therefore $\nabla_W L = 0$ throughout the unregularized flow. No distributional or zero-gain premise on W is needed. Gradient flow for the remaining parameters gives $\dot{b}_a = g_a c$ and $\dot{c} = \sum_b g_b b_b$. Differentiating $M_a = c \cdot b_a$, $u_{ab} = b_a \cdot b_b$, and $w = c \cdot c$ proves Eq. (55). The mode gradient has one component in the c block and one in its own b_a block, yielding $G_{ab} = u_{ab} + \delta_{ab} w$.

At initialization, the variances of M_s, M_w, u_{sw} are $1/N$ and those of $u_{ss} - 1, u_{ww} - 1, w - 1$ are $2/N$. Coordinatewise Chebyshev bounds and a union bound sum to $1 + 1 + 1 + 2 + 2 + 2 = 9$, proving Eq. (57).

The field bound $\|g(M)\|_\infty \leq A$ follows from $0 \leq \sigma \leq 1$. Since σ is $1/4$ -Lipschitz,

$$|g_a(M) - g_a(\tilde{M})| \leq \frac{1}{4} \sum_b \mathbb{E}|z_a z_b| \|M - \tilde{M}\|_\infty, \quad (89)$$

which proves Eq. (59). Define the Gram energy $R = w + u_{ss} + u_{ww}$. Along the deterministic flow,

$$R' = 4g \cdot M = 4\mathbb{E}[X\sigma(-X)], \quad X = z \cdot M. \quad (90)$$

Because $x\sigma(-x) \leq c_\star = W(e^{-1})$ and $R(0) = 3$, $R(t) \leq R_H$. Every coordinate of q is a squared norm or inner product of b_s, b_w, c , so Cauchy-Schwarz gives $\|q(t)\|_\infty \leq R(t) \leq R_H$.

Consider two states in the cube $\|q\|_\infty, \|\tilde{q}\|_\infty \leq B_{H,r}$ and set $d = \|q - \tilde{q}\|_\infty$. Every product in the M and u equations satisfies the direct estimate

$$|g_a(M)y - g_a(\tilde{M})\tilde{y}| \leq (A + B_{H,r}C)d. \quad (91)$$

There are at most three such products in an M equation and two in a u equation; the factor two and two cue indices give four in the w equation. Thus the full vector field is $L_{H,r} = 4(A + B_{H,r}C)$ -Lipschitz on the cube and therefore on the radius- r reference tube.

Let T_r be the first time before H that $\|q_N(t) - q(t)\|_\infty$ reaches the tube boundary. Up to T_r , Gronwall's inequality and Eq. (61) give

$$\|q_N(t) - q(t)\|_\infty \leq e^{L_{H,r}t} \|q_N(0) - q_0\|_\infty. \quad (92)$$

If $L_{H,r} > 0$, the initialization event makes this bound strictly less than r for every $t < H$, so the boundary cannot be reached before H . If $L_{H,r} = 0$, then $A = C = 0$, the cue law is zero almost surely, and the six-scalar vector field vanishes; the initial discrepancy is constant and cannot exit the closed tube. The endpoint may lie on the boundary. In either case the same estimate is at most δ throughout $[0, H]$, proving Eq. (62). Applying Eq. (57) with $\epsilon = e^{-L_{H,r}H} \min\{r, \delta\}$ proves Eq. (63). For the shared-initialization pair, use a common r and the maximum deterministic constant from the two fixed cue laws; the same concentrating initial six scalars control both paths. $\square$

K.10 PROOF OF COROLLARY 17

Under the stated noiseless positive rank-one cue law, the recurrence-invisible six-scalar equations give equal initial weak drifts in the two conditions for every finite realized W . Differentiating those drifts once at the symmetric state gives

$$(\dot{M}_w^B)'(0) = -\frac{\rho^2 + 1}{2}, \quad (\dot{M}_w^W)'(0) = -\frac{1}{2}, \quad (93)$$

so $\bar{d}'(0) = -\rho^2/2$. Taylor's theorem and continuity then make $\bar{\Delta}(t) < 0$ for all sufficiently small positive t .

The weak-only symmetric subsystem preserves $u_{ww} = w$. Direct differentiation shows $u_{ww}^2 - M_w^2 = 1$, yielding $u_{ww} = w = \sqrt{1 + M_w^2}$ and the displayed positive drift. While $0 \leq M_w \leq \beta$, it obeys $\dot{M}_w \geq 2\sigma(-\beta)$; integration proves Eq. (66). The finite-width statement is then exactly Theorem 16 applied at deterministic suppression and target-attainment witness times with margins smaller than both strict witness margins. $\square$

L PROOFS FOR LOCAL COUNTERFACTUAL DRIFT CORRECTION

L.1 INSTANTANEOUS PRESERVATION

If $a = 0$, both strong drifts vanish. Otherwise,

$$a \cdot q = a \cdot p - \frac{p \cdot a}{\|a\|^2} (a \cdot a) = a \cdot p - p \cdot a = 0, \quad (94)$$

so $a \cdot v' = a \cdot v + \alpha a \cdot q = a \cdot v$. This argument is independent of α , hence survives a cap. $\square$

L.2 UNIQUE MINIMUM-NORM CORRECTION

Let $\bar{\delta} = F_w^W - F_w$ be the raw deficit and $\delta = [\bar{\delta}]_+$. If $\bar{\delta} \leq 0$, $d = 0$ is feasible and uniquely norm-minimal. Suppose $\bar{\delta} > 0$, so $\delta = \bar{\delta}$. Any feasible d obeys $a \cdot d = 0$ and $p \cdot d \geq \delta$. Because $p = q + (p \cdot a / \|a\|^2)a$ when $a \neq 0$ (and $q = p$ when $a = 0$), $p \cdot d = q \cdot d$. Cauchy–Schwarz in the fixed Euclidean subspace orthogonal to a gives

$$\delta \leq q \cdot d \leq \|q\| \|d\|, \quad \|d\| \geq \delta / \|q\|. \quad (95)$$

Equality requires a nonnegative multiple of q . The correction $d^* = (\delta / \|q\|^2)q$ is feasible and attains this lower bound, so it is the unique minimum-norm, lower-bound-attaining feasible correction. Larger positive multiples can also attain the target but have larger norm. A positive deficit is feasible exactly when $q \neq 0$ in this representation. A nonbinding cap leaves d^* unchanged; a binding cap generally loses target attainment, so the target-attaining optimization claim no longer applies. $\square$

L.3 LOCAL FINITE-STEP DEVIATION

In the fixed chart and norm, L -Lipschitzness of the gradient on both straight segments gives the smoothness remainder

$$|m_s(\theta + h) - m_s(\theta) - \nabla m_s(\theta) \cdot h| \leq \frac{L}{2} \|h\|^2. \quad (96)$$

Apply this at $h = \eta v$ and $h = \eta v'$. Instantaneous preservation cancels the linear terms and the triangle inequality yields Eq. (69) for the actual velocities, capped or uncapped. For active uncapped CDC,

$$\|v'\| \leq \|v\| + \frac{\delta}{\|q\|} \leq V + D/q_{\min}, \quad (97)$$

which gives the explicit bound; a nonnegative upper cap can only decrease the added multiple of q . If smoothness is known only on the fixed-coordinate ball $B(\theta, r)$, sufficient segment conditions are $\eta V \leq r$ and $\eta(V + D/q_{\min}) \leq r$. These quantities and L are chart- and metric-dependent. $\square$

M DISCRETE CERTIFICATE AND EXPERIMENT AUDIT

For logged values Δ_k , increments telescope exactly:

$$\Delta_K - \Delta_0 = \sum_{k=0}^{K-1} (\Delta_{k+1} - \Delta_k). \quad (98)$$

The executable certificate requires one strict block of positive increments followed by one strict block of negative increments. The peak-minus-initial value is its positive area; peak-minus-final is its negative tail area. It certifies causal starvation only if the final trajectory is strictly below the initial gap and the external weak-only learnability flag is true. A false learnability flag blocks the causal certificate but does not establish that starvation is absent; status is indeterminate under that gate. This finite-step certificate makes no assertion between logged points.

Excluded engineering outcomes. Data seed 100 with model seeds 1000–1007 is excluded from scientific evidence. A first frozen-prediction grid mismatch stopped before training; a second observed- trajectory grid mismatch was detected after nonlinear outcomes had been generated, but no scores were inspected. The set was used only to repair evaluator alignment through integer-step matching plus a numerical optimization-time check. It did not change labels, thresholds, or the five E19 endpoints; E19 and E20 use fresh seeds.

Table 7: Executable contracts in `src/gradient_starvation/theory.py`.

Contract	Mathematical object	Scope guard
<code>projected_statistics</code>	direct Eq. (10), Gg , residual	direct drift primary for nonlinear probes
<code>initial_frozen_empirical_kernel</code>	J_0, Θ_0, c_0 in Corollary 3	initialization only; full sample kernel
<code>integrate_frozen_logistic_sgd</code>	explicit Euler for Eq. (15)	exact for tangent surrogate, not nonlinear model
<code>frozen_kernel_discrete_error_bound</code>	Eq. (22)	caller supplies secant-kernel movement bounds
<code>equal_time_drift_difference</code>	Δ' and two projected product orderings	exposes non-canonical attribution
<code>dense_linear_initial_gap_certificate</code>	Eqs. (30)–(31)	noiseless positive rank-one dense-linear scope only
<code>parameter_tube_certificate</code>	Theorem 10, Corollary 11	caller supplies uniform product-ball constants; missing premises fail closed
<code>rank_one_drift_ratio</code>	Theorem 12 factors and Ψ	caller supplies noiseless/exact hypotheses
<code>discrete_crossover_certificate</code>	telescoping finite-step tail areas	learnability gate required for causal label
<code>transverse_hitting_time_error_bound</code>	$\epsilon/\kappa + h$	supplied error, prehistory, radius, and mesh guards
<code>zero_disorder_failure_bound</code>	Eqs. (57), (63)	caller supplies reference-tube constants
<code>cdc_finite_step_deviation_bound</code>	Eq. (69)	caller supplies local smoothness and segment containment

Table 8: Executed experiment configurations. “Shadow” methods train a second weak-only model and therefore use roughly twice the model memory and per-step compute.

Experiment	Model / width	Task grid	Samples	Steps / LR	Seeds / records	Primary purpose
E1	dense linear / 96	$\rho \in \{1, 2, 4, 8\}$, lag $\in \{0, 2, 4, 8\}$	256	500 / .01	4–11	paired deficit / gate audit
E2	dense linear / 32–256	$(1, 0), (2, 2), (4, 4), (8, 8)$	1024	400 / .01	0–7	empirical reference
E-NL	tanh, GRU / 32	$\rho = 4$, lag 2	256	1000 / .01	20–27	direct-drift certificate
Tangent pilot	tanh, GRU / 32	$\rho = 4$, lag 2	256	1000 / .01	1 × 8	five-output falsification
Tangent factorial (E20)	tanh, GRU / 32	$\rho = 4$, lag 2	256	1000 / .01	4 × 8 / model	primary crossing-event audit
Historical E3	dense linear / 96	$\rho = 4$, lag 2	256	500 / .01	40–47	CDC local ablation
E26	semi-real CNN	MNIST 3-vs-8; FashionMNIST 0-vs-6	512/256/256 per dataset	500 / .01	4 × 8 / dataset; 64	generated-cue intervention
E27	tanh, GRU / 32	three cells; $\beta \in \{.25, .5, 1, 2\}$	256	1000 / .01	4 × 8; 192	all- β robustness
E28	tanh / 64	two cells; ERM, CDC, Bloop-style shadow-target rescue, PCGrad-style shadow-target rescue	256	1000 / .01	4 × 8; 256 method-records	comparator-restricted CDC tradeoff

Supplementary artifact map and commands. The anonymous supplementary archive separates the controlled paired-deficit and gate audit, finite-width reference, exact solver checks, final nonlinear run and terminal β audit, failed broad tangent pilot, restricted E20 tangent factorial, exploratory negative search, historical E3 CDC ablation, E26 semi-real negative/indeterminate study, E27 beta-robust negative study, and E28 mixed CDC tradeoff. E20 remains submission-primary with only drift/response crossings primary. Machine-readable provenance binds each claim to configurations, outcomes, seals, and source status. Run `python verify_release_artifacts.py` for the read-only compact-archive audit. New scientific runs use `scripts/run_semi_real_study.py` or `scripts/run_expanded_studies.py` only after fresh preflight and external seal; they are not replays of omitted record bytes. E26–E28 use reconstructible clean source at commit `9ebbc617...2d3517d0`, but the E26 and combined E27/E28 compact archives omit 192 and 1,792 records/** files, respectively, so they are auditable rather than standalone replay packages. E19/E20 instead lack their historical dirty source bytes and are not historically replayable.

N COMPLETED CAUSAL-GATE AND MECHANISM-CONTROL OUTCOME AND PROVENANCE AUDIT

These studies were prospectively frozen, externally authorized, and executed from clean Git commit 9ebbc617a9aaad5c3c19e9d5fd4fa08f2d3517d0. They are later breadth studies, not an outcome-dependent replacement for E20: E20 remains the submission-primary empirical claim and only its drift- and response-crossing classification endpoints are primary.

N.1 E26: SEMI-REAL CAUSAL INDETERMINACY AFTER WEAK-ONLY GATE FAILURE

E26 contains 64 crossed records: four data seeds by eight model seeds for MNIST 3-vs-8 and the same factorial for FashionMNIST 0-vs-6. The core channel contains selected real benchmark pixels; the paired intervention exactly zeros only the generated cue channel. Thus the evidence label is semi-real generated-cue, not an unmodified real-world or natural-spurious-correlation benchmark. Positive weak AUC gap means the weak-only core gain exceeds the both-feature core gain.

Table 9: Complete E26 primary outcomes. CIs are dataset-specific crossed-seed intervals.

Dataset	Weak-only learnable	Causal certificate	Mean weak AUC gap (95% CI)
MNIST 3-vs-8	0/32	0/32	−0.0012362842136667493 [−0.0034405428466004646, 0.0010159591371180453]
FashionMNIST 0-vs-6	0/32	0/32	−0.00010921728159018996 [−0.004129384520886106, 0.003182670049955049]

The frozen hypothesis was not accepted. Because the weak-only first-hit gate failed in all 32 records per dataset, the causal gate rejects a false-positive starvation interpretation. The zero causal counts remain negative/indeterminate: they cannot be interpreted as evidence that starvation is absent or that either dataset is intrinsically unlearnable. The compact archive appears under `paper/artifacts/` as `semi-real-generated-cue-v1-20260830`; the central provenance manifest records its exact archive, artifact, and 22-file clean-source digests. Source is reconstructible from the clean commit, but the compact archive omits 192 `records/**` files; hashes bind those files but cannot recover their bytes, so the archive is not a standalone replay package.

N.2 E27: EXPANDED BETA-ROBUST NONLINEAR STUDY

E27 contains 192 records: tanh and GRU across three fixed task cells and a crossed 4×8 seed design. Each record evaluates the prospectively frozen first-hit grid $\beta \in \{0.25, 0.5, 1, 2\}$; the architecture estimand equally weights the three cell prevalences. A 5,000-replicate two-way pigeonhole bootstrap supplies raw percentile intervals, and the two one-sided architecture tests form one Holm family.

Table 10: Complete E27 all- β architecture endpoints.

Architecture	Macro estimate	95% CI	Holm-adjusted p
tanh	0.2916666666666667	[0.16666666666666666, 0.4166666666666667]	1.0
GRU	0	[0, 0]	1.0

The overall beta-robust claim is false. Architecture ranking was forbidden by the frozen contract and was neither performed nor supported.

N.3 E28: COMPARATOR-RESTRICTED MECHANISM-LEVEL CONTROL

E28 contains 256 method-records for ERM, CDC, Bloop-style shadow-target rescue, and PCGrad-style shadow-target rescue over two fixed cells and crossed seeds. Both comparators are non-canonical (`canonical:false`) response-gradient shadow-target variants; the study does not establish canonical implementation parity. Weak rescue is $-|\text{weak AUC gap}|$ and is higher-better, so CDC-minus-comparator positive values favor CDC. Trajectory deviation is absolute strong-response AUC deviation from paired ERM and final deviation is absolute final strong-response deviation from paired ERM; both are lower-better, so comparator-minus-CDC positive values favor CDC.

The frozen joint weak-rescue/strong-preservation tradeoff is true only against Bloop-style shadow-target rescue and false against PCGrad-style shadow-target rescue. CDC therefore does not beat the

Table 11: Complete E28 frozen equal-weight two-cell macro contrasts. Brackets are 95% two-way-bootstrap CIs.

Comparator	CDC-comparator weak rescue	Comparator-CDC trajectory deviation	Comparator-CDC final deviation	Joint tradeoff
Blisep-style shadow-target rescue	0.002198250399210656 [0.0009779187294890126, 0.003277366267916477]	3.4651361294978416 [3.3443827215311754, 3.5849715262780952]	0.0422362388111651 [0.0288088704226539, 0.056116884160315]	true
PCGrad-style shadow-target rescue	-0.00015301238407074821 [-0.00043769348592838795, 0.00013248012419808214]	-0.4870688912867656 [-0.6887378844428417, -0.2773782773489674]	-0.2565438437728021 [-0.3018841153814253, -0.2087651835774514]	false

latter under the frozen joint criterion. Because both comparators are non-canonical, E28 supports no SOTA, canonical parity, or broad CDC superiority. E27 and E28 share `paper/artifacts/expanded-studies-v1-20260830`; the central provenance manifest records its exact archive, artifact, and 23-file clean-source digests. Source is reconstructible from the clean commit, but the compact archive omits 1,792 `records/**` files and is not a standalone replay package.

O FROZEN-TANGENT FALSIFICATION AND RESTRICTED FOLLOW-UP

The sealed E19 pilot tested one preregistered conjunction over five outputs and 16 architecture-seed records. Correct classifications were phase 12/16, drift crossing 14/16, response crossing 14/16, causal certificate 14/16, and weak-only learnability 12/16. The conjunction failed because tanh learnability was 4/8, below the frozen 5/8 floor; GRU learnability was 8/8. Magnitude calibration also differed sharply by architecture: tanh response-gap, weak-response, and drift RMSE were 2.367, 2.394, and 0.486, versus 0.0163, 0.0281, and 0.0107 for GRU. These numbers are preserved as a falsification result rather than averaged into a pass.

E20 was specified only after E19’s rejection and used fresh crossed seeds. Its accepted scope was restricted to event labels: response crossing was correct in 64/64 records and drift crossing in 60/64 (32/32 tanh, 28/32 GRU), with two-way bootstrap intervals $[1, 1]$ and $[0.8125, 1]$. The rejected secondary outputs were substantially weaker: phase 52/64, causal certificate 56/64, and learnability 49/64, including tanh learnability 17/32. Predictions were systematically early (overall drift/response bias $-0.822/-1.189$), while tanh response-gap, weak-response, and drift RMSE were 2.455, 2.489, and 0.490. Consequently E20 is an uncalibrated crossing-event classification result at one controlled cell, not a phase, causal, learnability, time, magnitude, prevalence, or kernel-stability validation. Both compact archives preserve these scores and seals but not the unavailable exact dirty source snapshots, so integrity auditability does not imply historical replayability.